Operating Systems (Honor Track)

Memory 3: Demand Paging

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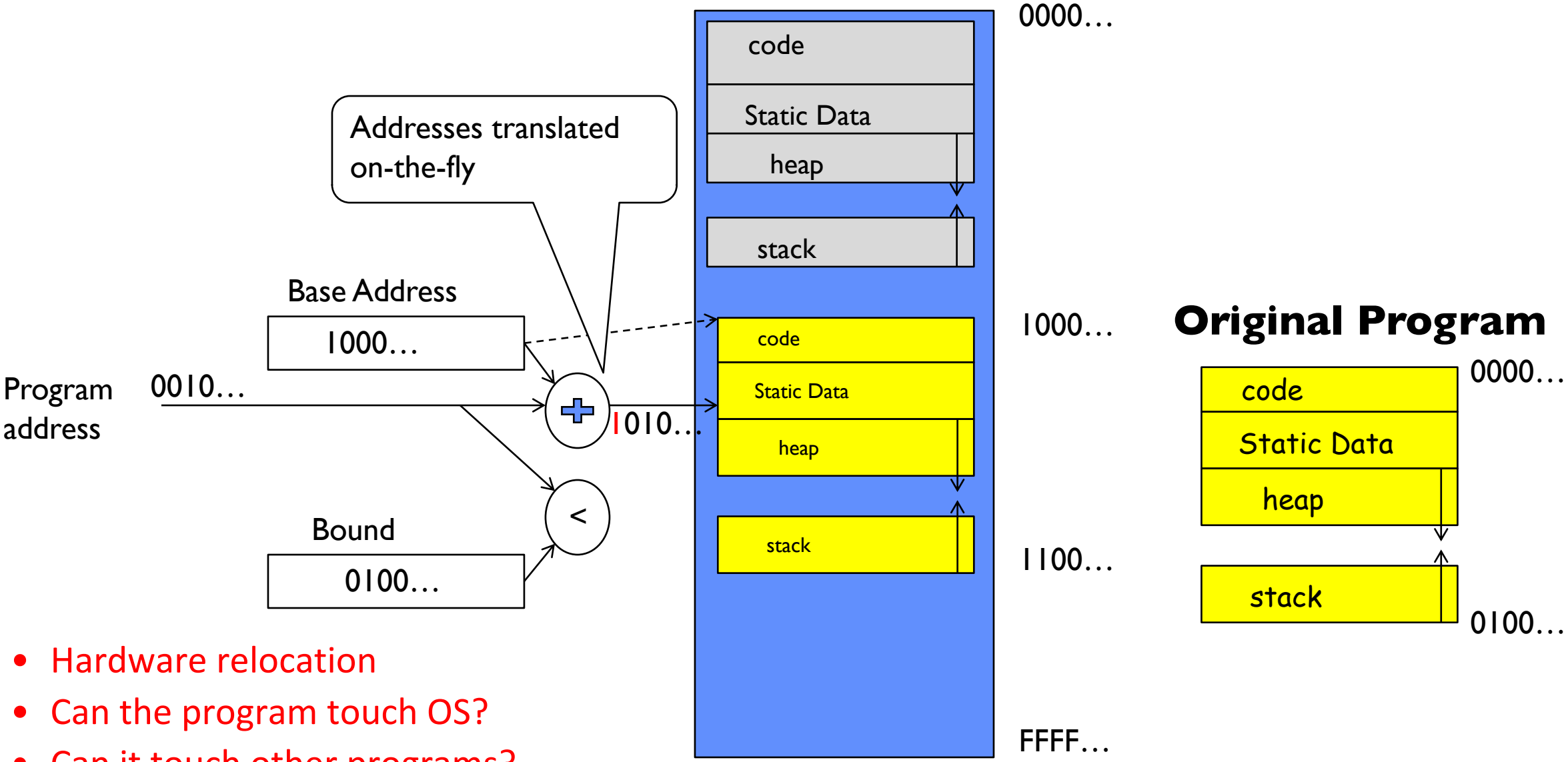
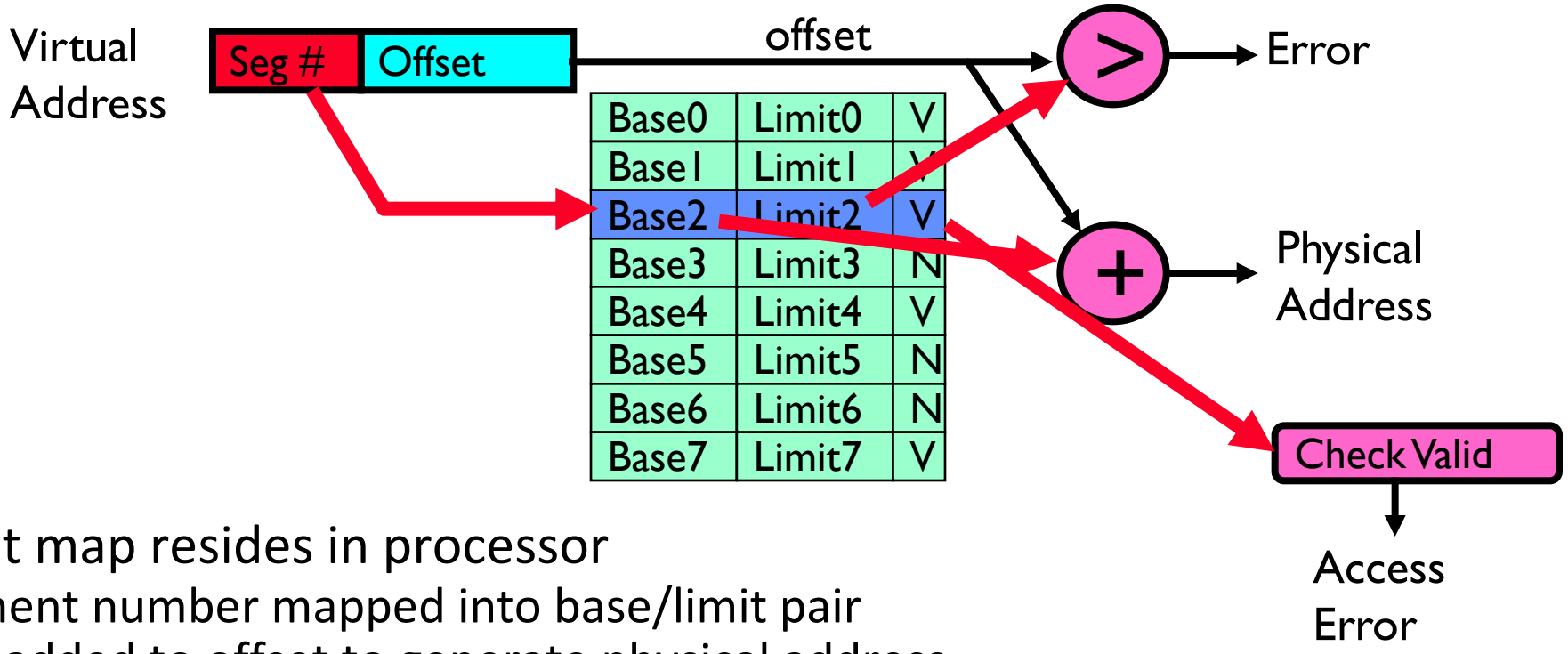
Acknowledgments: Ion Stoica, Berkeley CS 162

Recap: Base and Bound (with Translation)



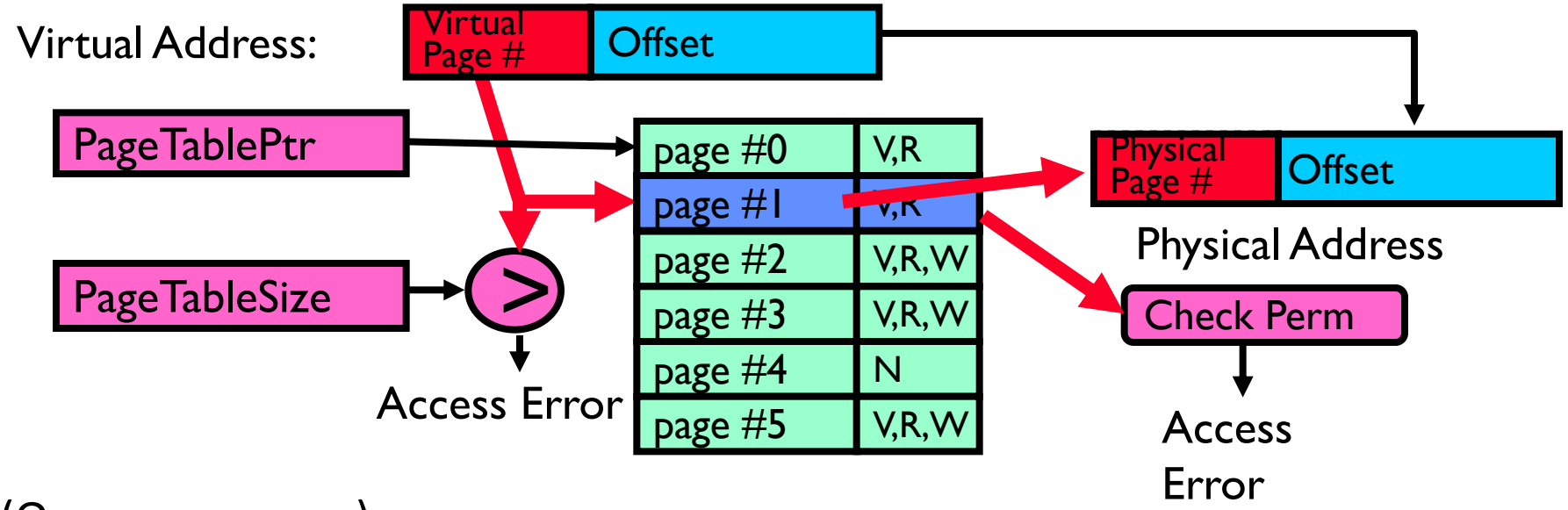
- Hardware relocation
- Can the program touch OS?
- Can it touch other programs?

Recap: Implementation of Multi-Segment Model



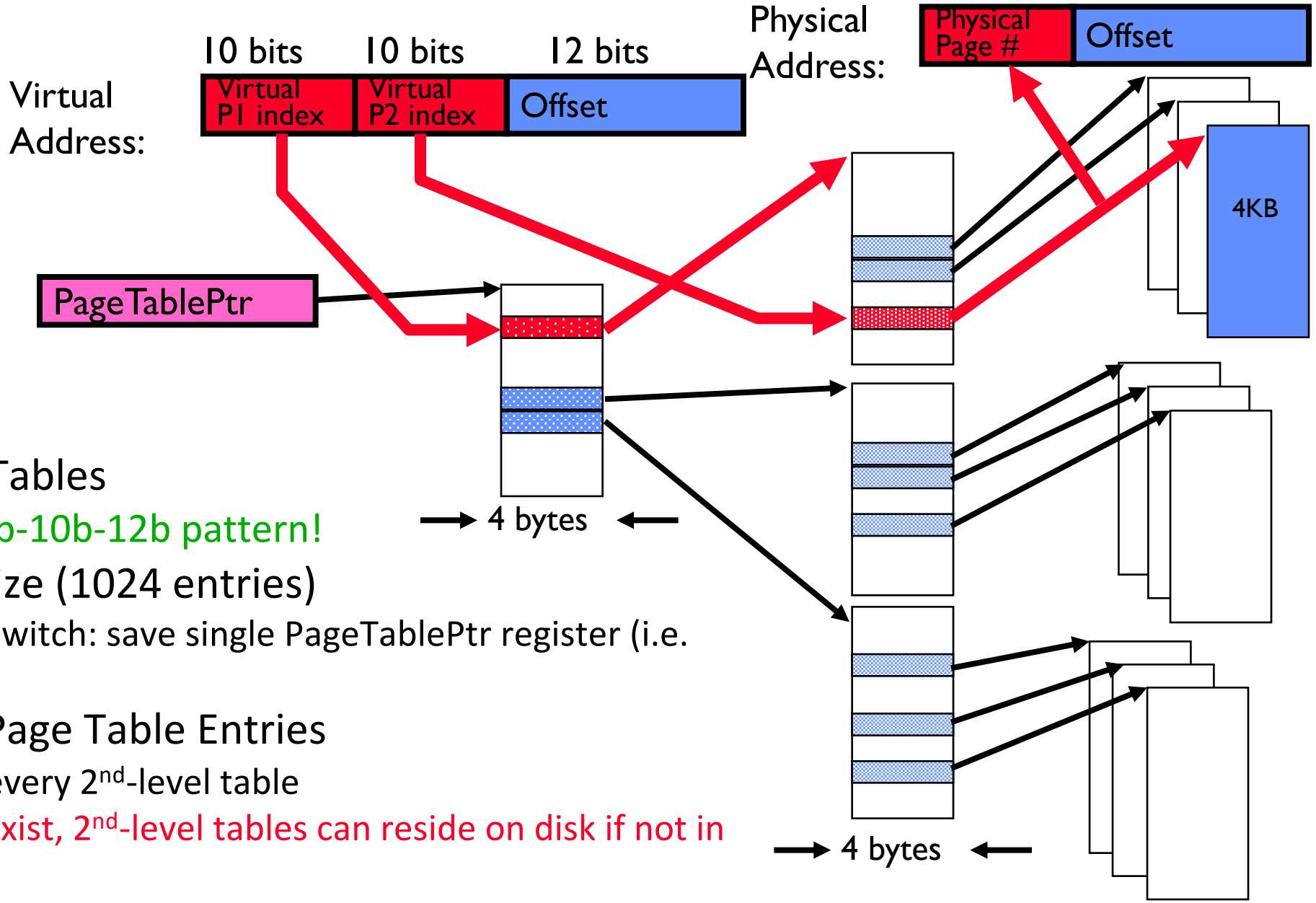
- Segment map resides in processor
  - Segment number mapped into base/limit pair
  - Base added to offset to generate physical address
  - Error check catches offset out of range
- As many chunks of physical memory as entries
  - Segment addressed by portion of virtual address
  - However, could be included in instruction instead:
    - » x86 Example: mov [es:bx],ax.
- What is "V/N" (valid / not valid)?
  - Can mark segments as invalid; requires check as well

Recap: How to Implement Simple Paging?



- Page Table (One per process)
  - Resides in physical memory
  - Contains physical page and permission for each virtual page (e.g. Valid bits, Read, Write, etc.)
- Virtual address mapping
  - Offset from Virtual address copied to Physical Address
    - » Example: 10 bit offset ⇒ 1024-byte pages
  - Virtual page # is all remaining bits
    - » Example for 32-bits: 32-10 = 22 bits, i.e. 4 million entries
    - » Physical page # copied from table into physical address
  - Check Page Table bounds and permissions

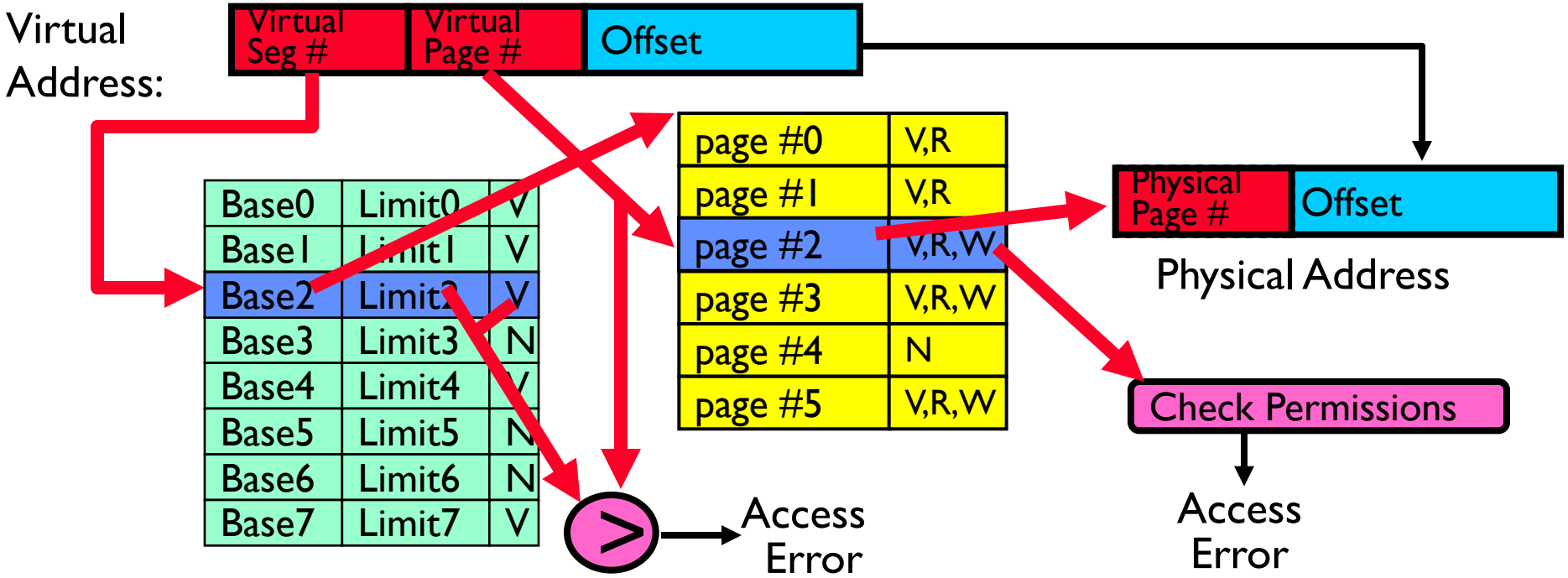
Recap: The two-level page table



- Tree of Page Tables
  - "Magic" 10b-10b-12b pattern!
- Tables fixed size (1024 entries)
  - On context-switch: save single PageTablePtr register (i.e. CR3)
- Valid bits on Page Table Entries
  - Don't need every 2nd-level table
  - Even when exist, 2nd-level tables can reside on disk if not in use

Recap: Multi-level Translation: Segments + Pages

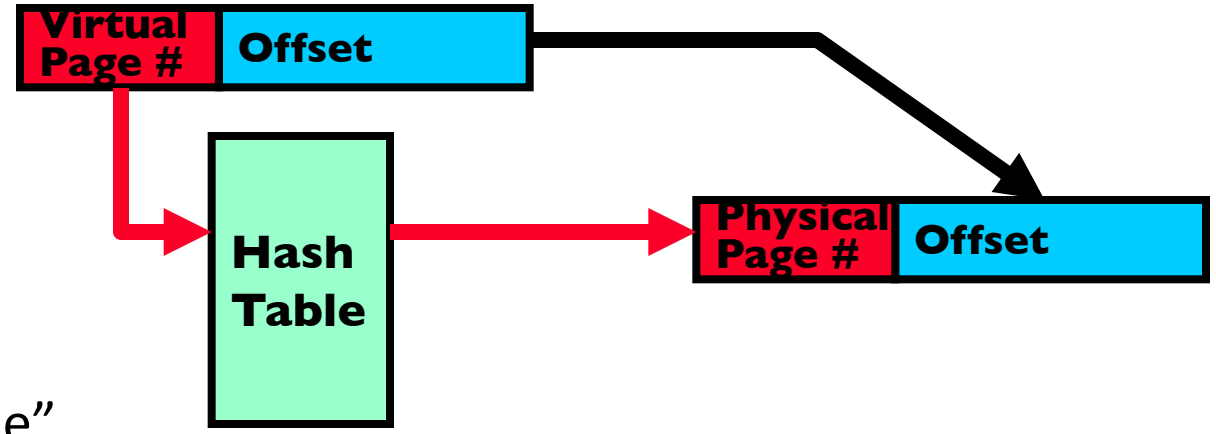
- What about a tree of tables?
  - Lowest level page table ⇒ memory still allocated with bitmap
  - Higher levels often segmented
- Could have any number of levels. Example (top segment):



- What must be saved/restored on context switch?
  - Contents of top-level segment registers (for this example)
  - Pointer to top-level table (page table)

Recap: Inverted Page Table

- With all previous examples ("Forward Page Tables")
  - Size of page table is at least as large as amount of virtual memory allocated to processes
  - Physical memory may be much less
    - » Much of process space may be out on disk or not in use



- Answer: use a hash table
  - Called an "Inverted Page Table"
  - Size is independent of virtual address s
  - Directly related to amount of physical
  - Very attractive option for 64-bit addre
    - » PowerPC, UltraSPARC, IA64

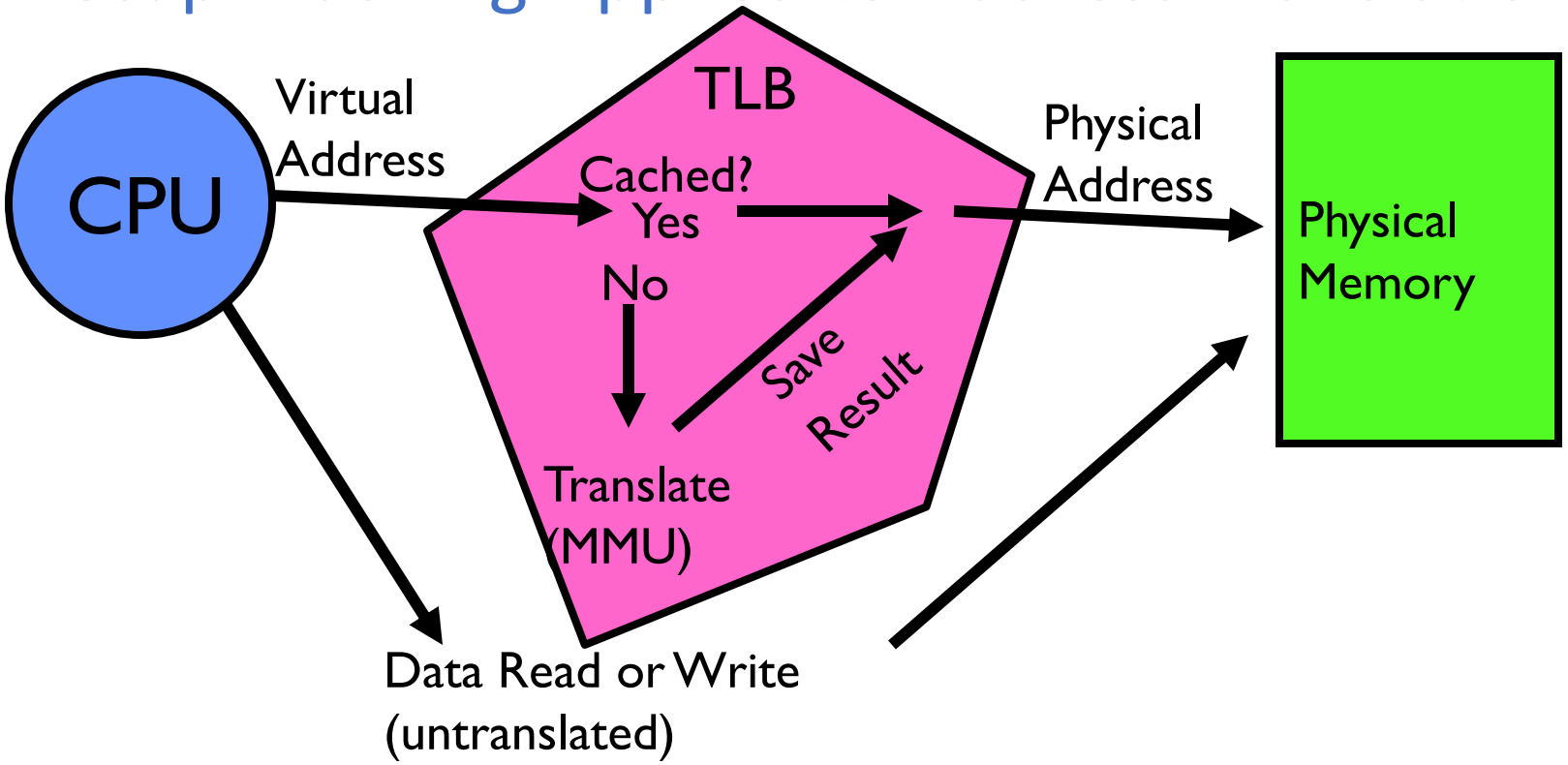
Total size of page table ≈ number of pages used by program in physical memory. Hash more complex

- Cons:
  - Complexity of managing hash chains: Often in hardware!
  - Poor cache locality of page table

Recap: Address Translation Comparison

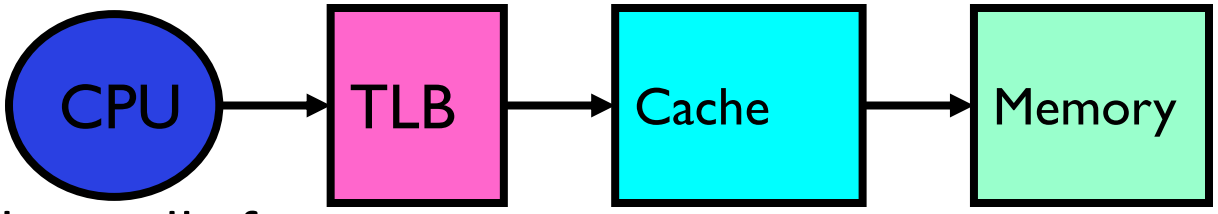
	Advantages	Disadvantages
Simple Segmentation	Fast context switching (segment map maintained by CPU)	Internal/External fragmentation
Paging (Single-Level)	No external fragmentation Fast and easy allocation	Large table size (~ virtual memory) Internal fragmentation
Paged Segmentation	Table size ~ # of pages in virtual memory Fast and easy allocation	Multiple memory references per page access
Multi-Level Paging		
Inverted Page Table	Table size ~ # of pages in physical memory	Hash function more complex No cache locality of page table

Recap: Caching Applied to Address Translation



- Question is one of page locality: does it exist?
  - Instruction accesses spend a lot of time on the same page (since accesses sequential)
  - Stack accesses have definite locality of reference
  - Data accesses have less page locality, but still some…
- Can we have a TLB hierarchy?
  - Sure: multiple levels at different sizes/speeds

What TLB Organization Makes Sense?



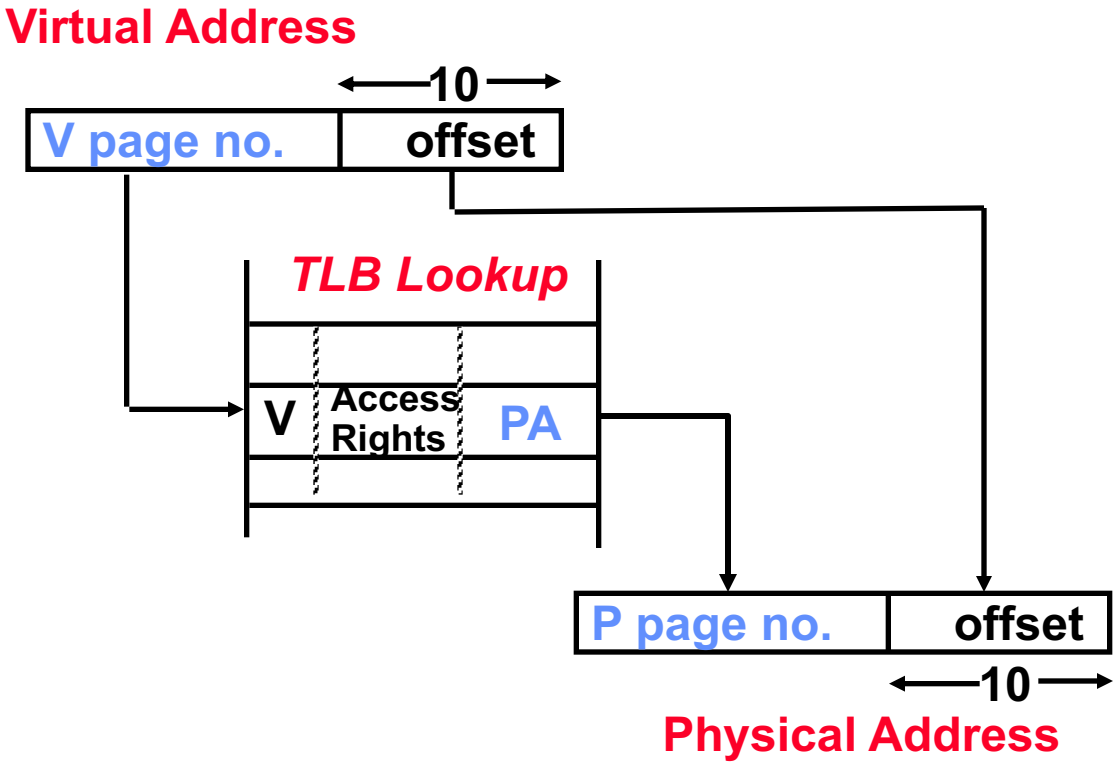
- Needs to be really fast
  - Critical path of memory access
    - » In simplest view: before the cache
    - » Thus, this adds to access time (reducing cache speed)
  - Seems to argue for Direct Mapped or Low Associativity
- However, needs to have very few conflicts!
  - With TLB, the Miss Time extremely high! (PT traversal)
  - Cost of Conflict (Miss Time) is high
  - Hit Time – dictated by clock cycle
- Thrashing: continuous conflicts between accesses
  - What if use low order bits of page as index into TLB?
    - » First page of code, data, stack may map to same entry
    - » Need 3-way associativity at least?
  - What if use high order bits as index?
    - » TLB mostly unused for small programs

TLB organization: include protection

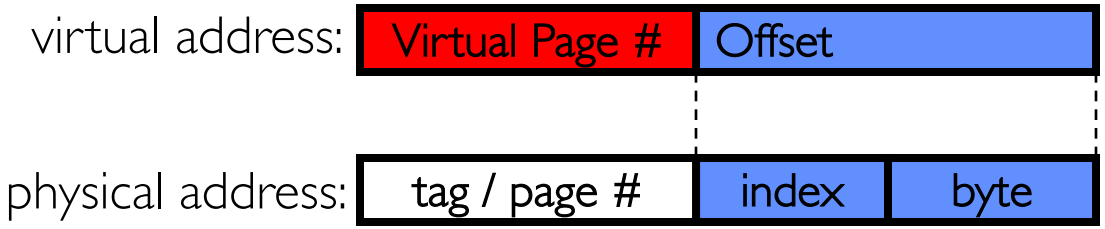
- How big does TLB actually have to be?
  - Usually small: 128-512 entries (larger now)
  - Not very big, can support higher associativity
- Small TLBs usually organized as fully-associative cache
  - Lookup is by Virtual Address
  - Returns Physical Address + other info
- What happens when fully-associative is too slow?
  - Put a small (4-16 entry) direct-mapped cache in front
  - Called a “TLB Slice”

Reducing translation time for physically-indexed caches

- As described, TLB lookup is in serial with cache lookup
  - Consequently, speed of TLB can impact speed of access to cache

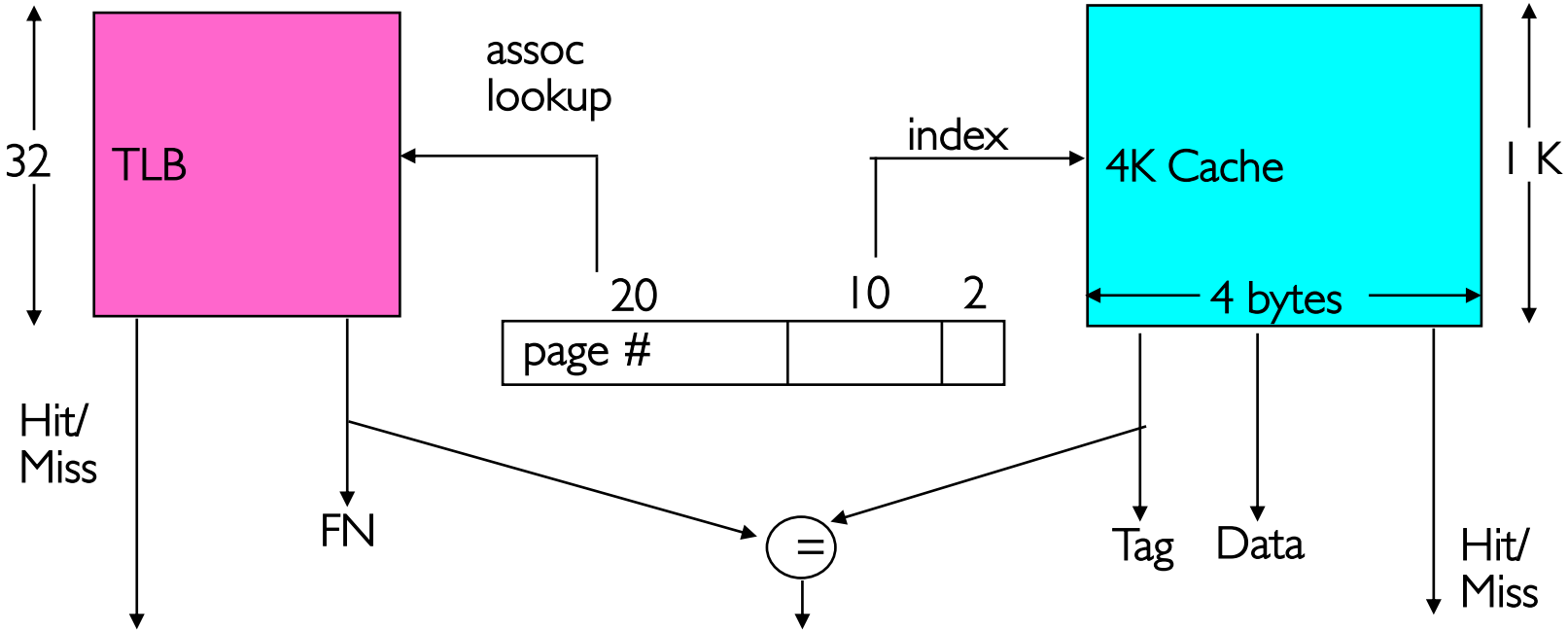


- Machines with TLBs go one step further: overlap TLB lookup with cache access
  - Works because offset available early
  - Offset in virtual address exactly covers the "cache index" and "byte select"
  - Thus can select the cached byte(s) in parallel to perform address translation



Overlapping TLB & Cache Access

- Here is how this might work with a 4K cache:

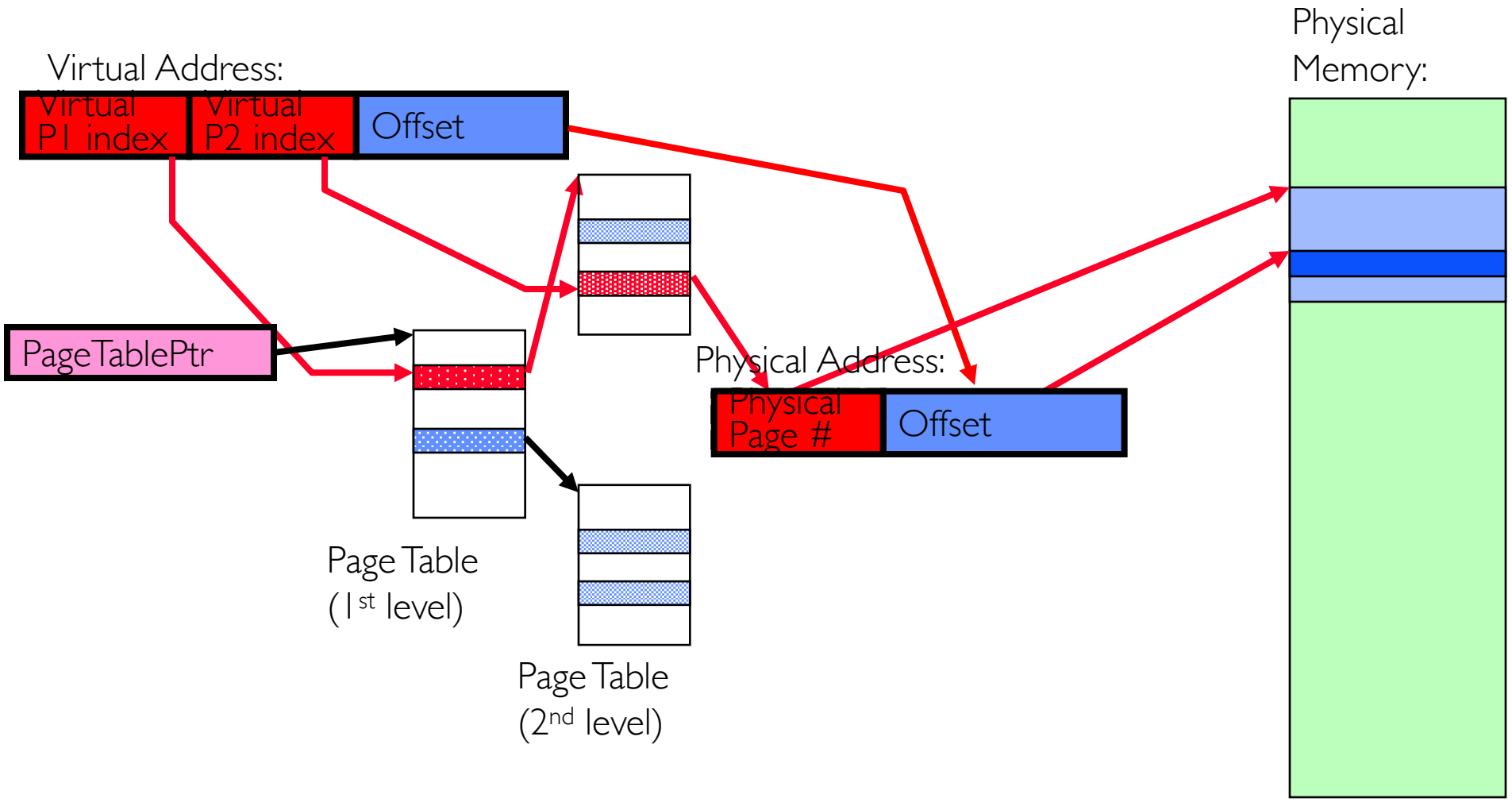


- What if cache size is increased to 8KB?
  - Overlap not complete
  - Need to do something else
- Another option: Virtual Caches would make this faster
  - Tags in cache are virtual addresses
  - Translation only happens on cache misses

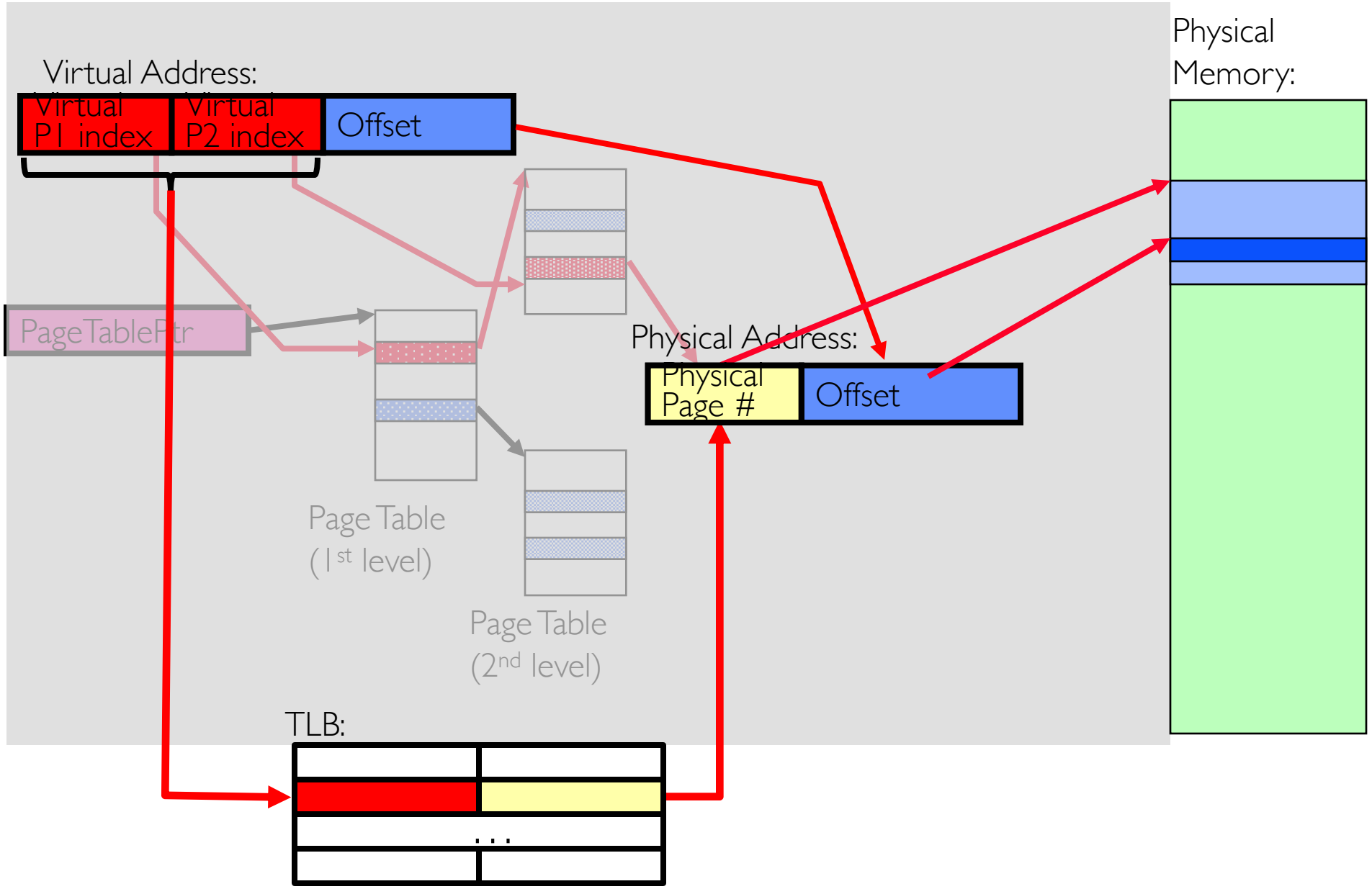
What happens on a Context Switch?

- Need to do something, since TLBs map virtual addresses to physical addresses
  - Address Space just changed, so TLB entries no longer valid!
- Options?
  - Invalidate TLB: simple but might be expensive
    - » What if switching frequently between processes?
  - Include ProcessID in TLB
    - » This is an architectural solution: needs hardware
- What if translation tables change?
  - For example, to move page from memory to disk or vice versa…
  - Must invalidate TLB entry!
    - » Otherwise, might think that page is still in memory!
  - Called “TLB Consistency”
- Aside: with Virtually-Indexed cache, need to flush cache!
  - Remember, everyone has their own version of the address “0”!

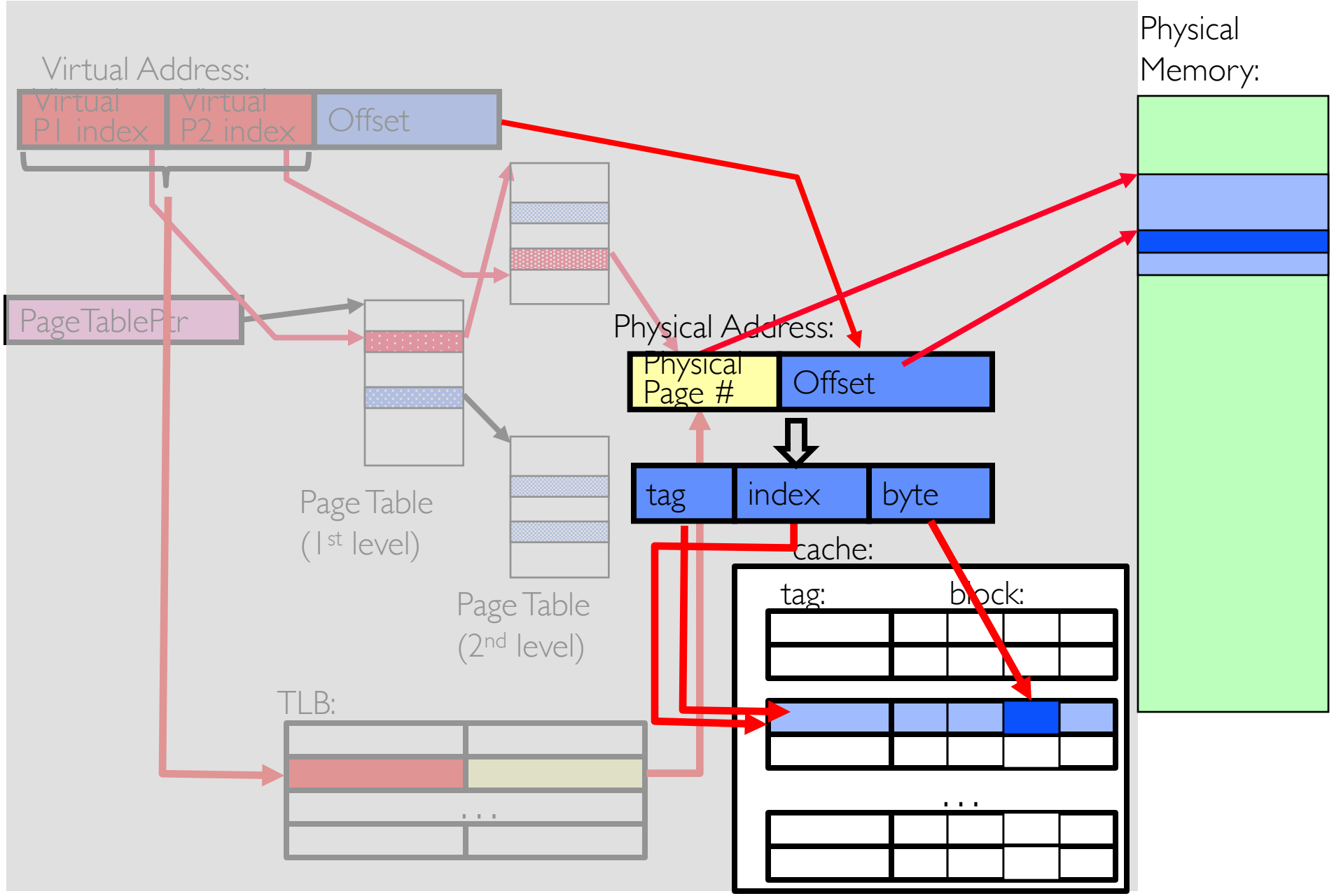
Putting Everything Together: Address Translation



Putting Everything Together: TLB



Putting Everything Together: Cache

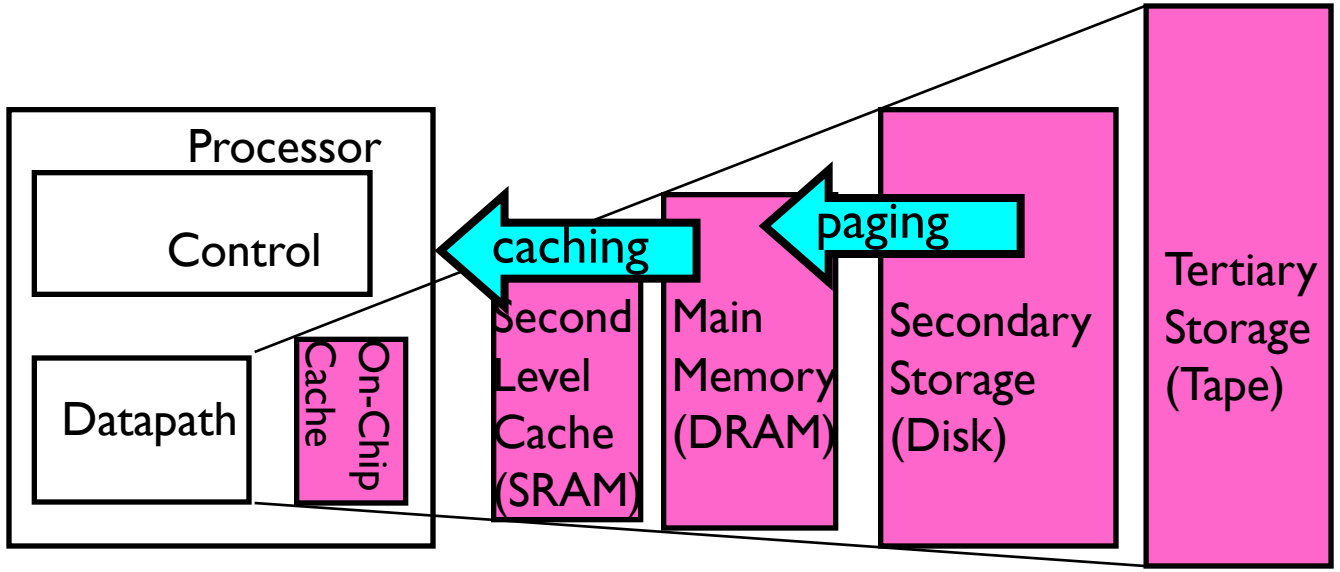


Page Fault

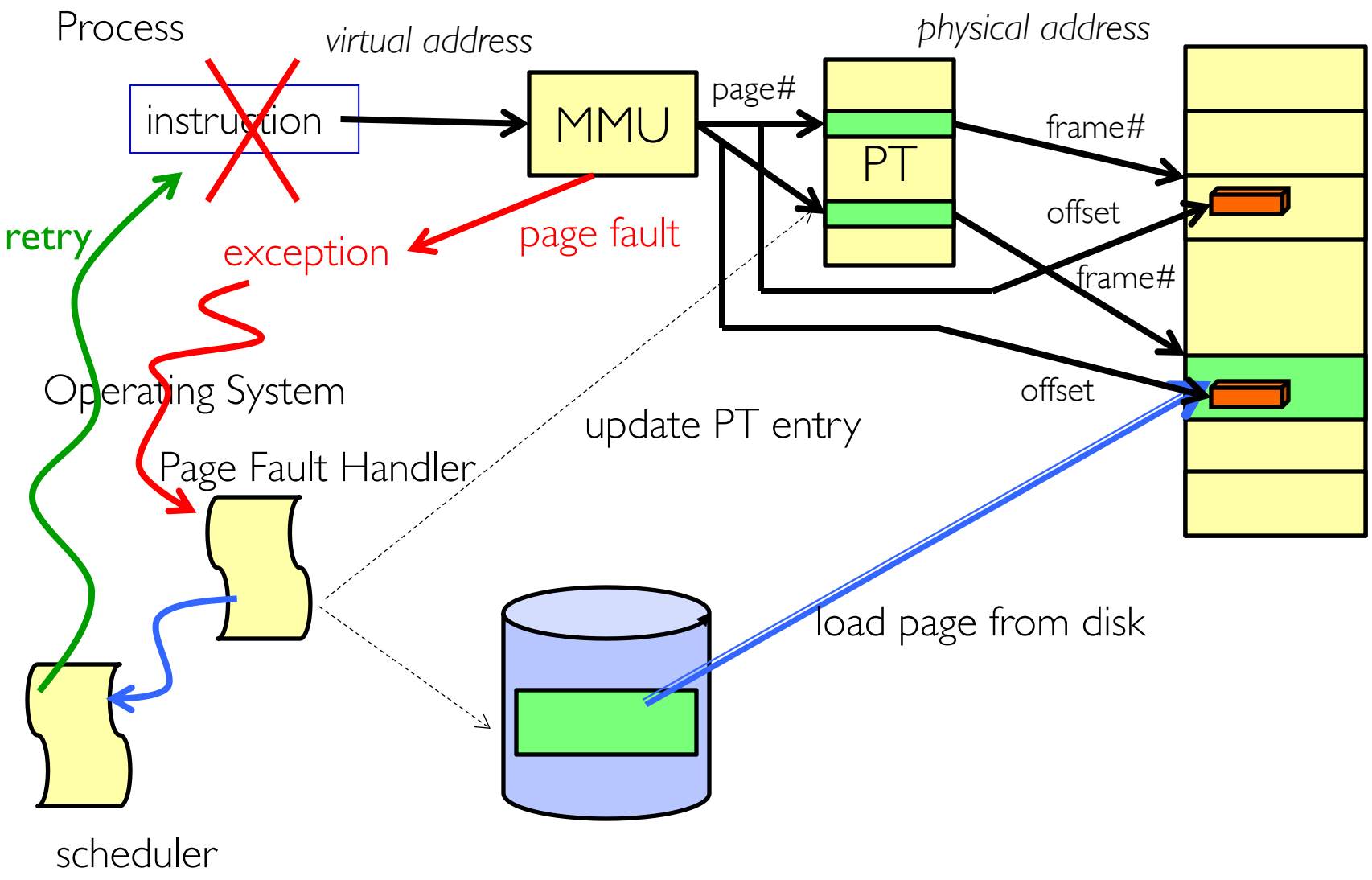
- The Virtual-to-Physical Translation fails
  - PTE marked invalid, Privilege Level Violation, Access violation, or does not exist
  - Causes a Fault / Trap
    - » Not an interrupt because synchronous to instruction execution
  - May occur on instruction fetch or data access
- Protection violations typically terminate the instruction
- Other Page Faults engage operating system to fix the situation and retry the instruction
  - Allocate an additional stack page, or
  - Make the page accessible - Copy on Write,
  - Bring page in from secondary storage to memory – demand paging
- Fundamental inversion of the hardware / software boundary

Demand Paging

- Modern programs require a lot of physical memory
  - Memory per system growing faster than 25%-30%/year
- But they don't use all their memory all of the time
  - 90-10 rule: programs spend 90% of their time in 10% of their code
  - Wasteful to require all of user's code to be in memory
- Solution: use main memory as "cache" for disk



Page Fault ⇒ Demand Paging



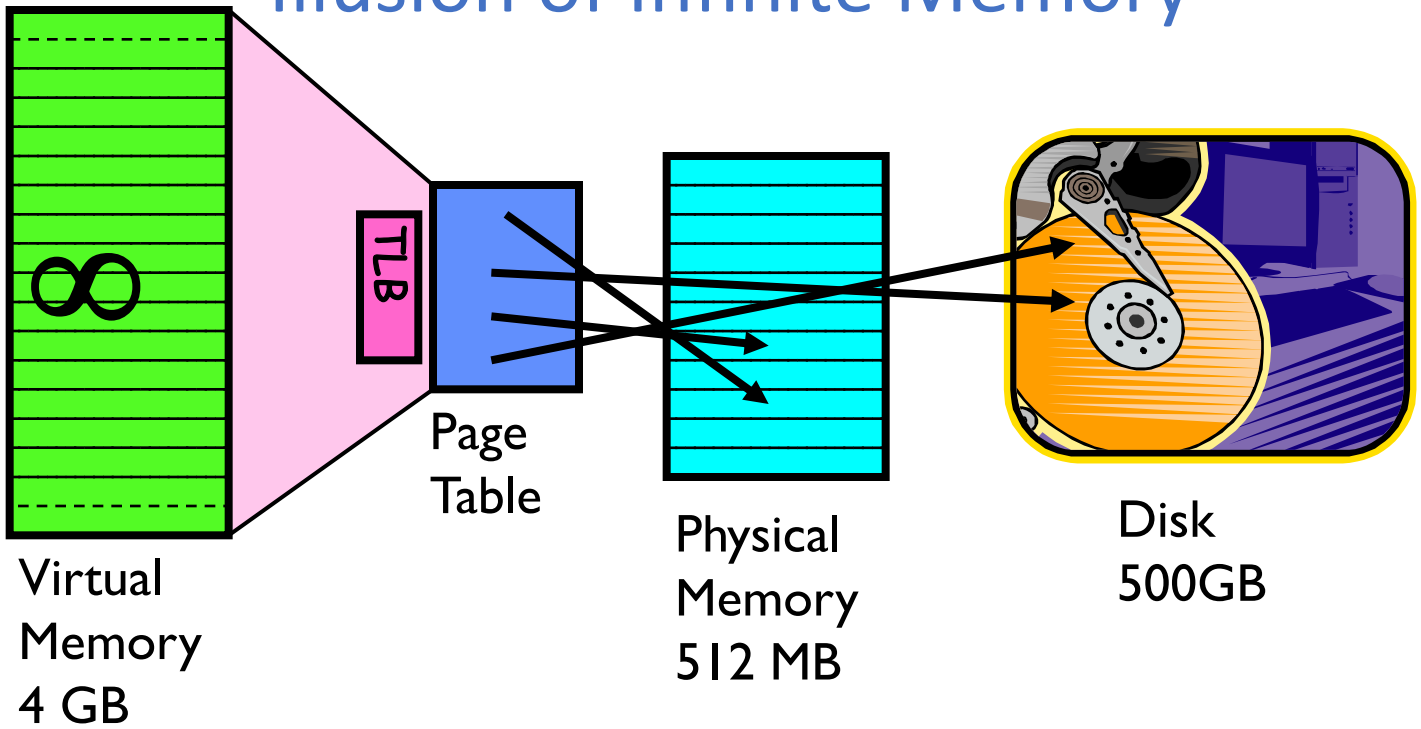
Group Discussion: Demand Paging as Caching, …

- What “block size”?
- What “organization” i.e., direct-mapped, set-associative, fully-associative?
- How do we locate a page?
- What is page replacement policy? (i.e., LRU, Random…)
- What happens on a miss?
- What happens on a write? (write-through, write back)

Demand Paging as Caching, …

- What "block size"? - 1 page (e.g., 4 KB)
- What "organization" i.e., direct-mapped, set-associative, fully-associative?
  - Fully associative since arbitrary mapping
- How do we locate a page?
  - First check TLB, then page-table traversal
- What is page replacement policy? (i.e., LRU, Random…)
  - This requires more explanation… (more later)
- What happens on a miss?
  - Go to lower level to fill miss (i.e., disk)
- What happens on a write? (write-through, write back)
  - Definitely write-back – need dirty bit!

Illusion of Infinite Memory

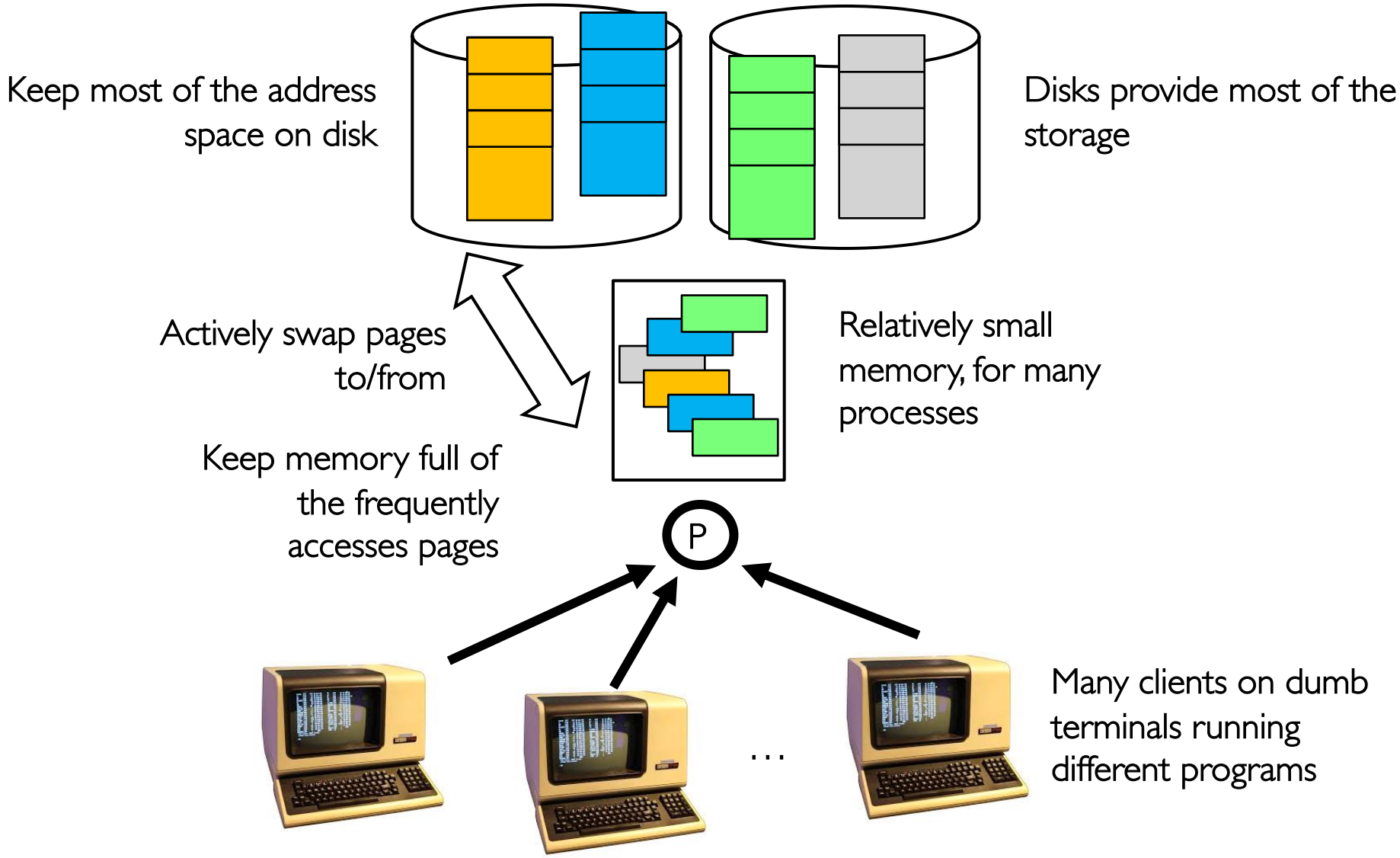


- Disk is larger than physical memory ⇒
  - In-use virtual memory can be bigger than physical memory
  - Combined memory of running processes much larger than physical memory
    - » More programs fit into memory, allowing more concurrency
- Principle: Transparent Level of Indirection (page table)
  - Supports flexible placement of physical data
    - » Data could be on disk or somewhere across network (NSDI'17 InfiniSwap, OSDI'20 AIFM)
  - Variable location of data transparent to user program
    - » Performance issue, not correctness issue

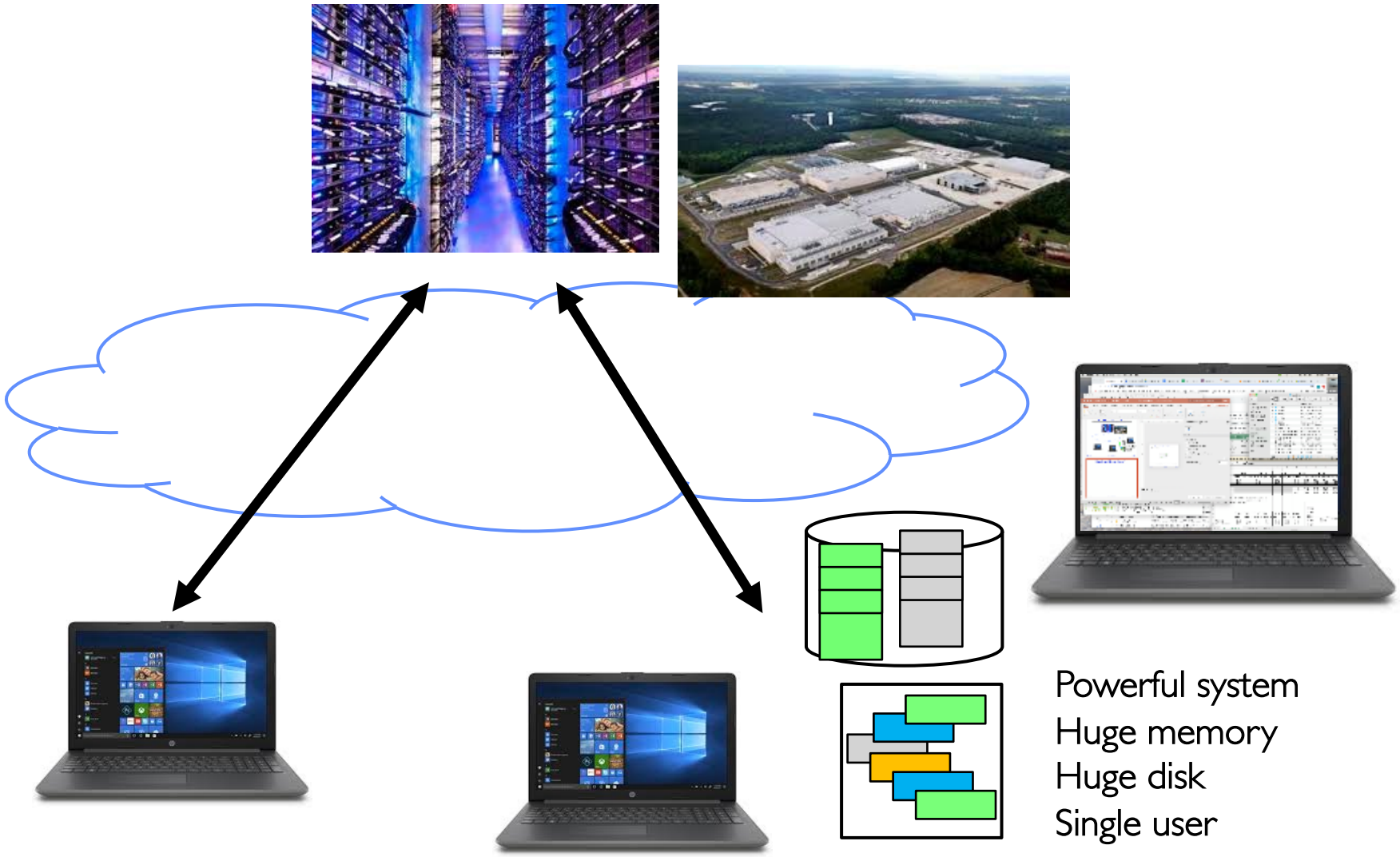
Demand Paging Mechanisms

- PTE makes demand paging implementable
  - Valid ⇒ Page in memory, PTE points at physical page
  - Not Valid ⇒ Page not in memory; use info in PTE to find it on disk when necessary
- Suppose user references page with invalid PTE?
  - Memory Management Unit (MMU) traps to OS
    - » Resulting trap is a "Page Fault"
  - What does OS do on a Page Fault?:
    - » Choose an old page to replace
    - » If old page modified ("Dirty=1"), write contents back to disk
    - » Change its PTE and any cached TLB to be invalid
    - » Load new page into memory from disk
    - » Update page table entry
    - » Continue thread from original faulting location
  - TLB for new page will be loaded when thread continued!
  - While pulling pages off disk for one process, OS runs another process from ready queue
    - » Suspended process sits on wait queue

Origins of Paging



Very Different Situation Today



A Picture on one machine

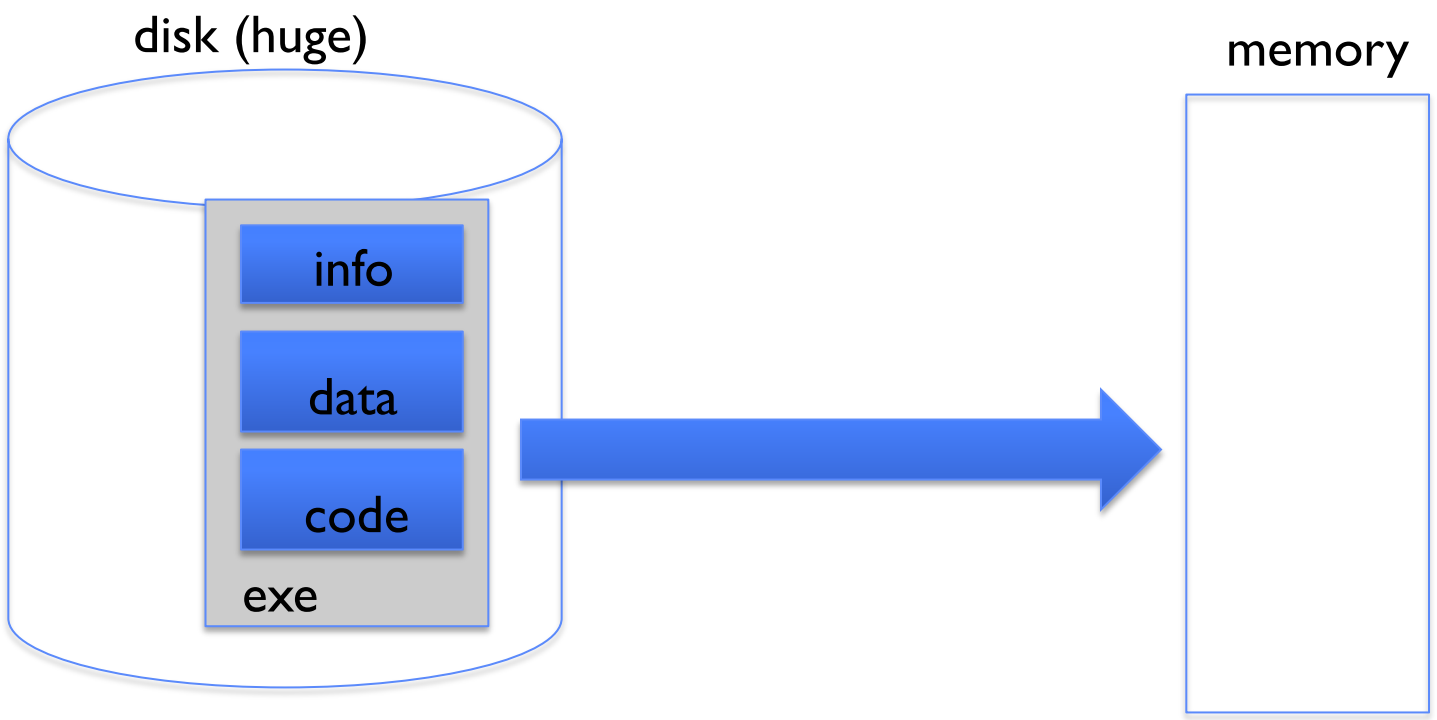
```
Processes: 407 total, 2 running, 405 sleeping, 2135 threads                    22:10:39
Load Avg: 1.26, 1.26, 0.98  CPU usage: 1.35% user, 1.59% sys, 97.5% idle
SharedLibs: 292M resident, 54M data, 43M linkedit.
MemRegions: 155071 total, 4489M resident, 124M private, 1891M shared.
PhysMem: 13G used (3518M wired), 2718M unused.
VM: 1819G vsize, 1372M framework vsize, 68020510(0) swapins, 71200340(0) swapouts.
Networks: packets: 40629441/21G in, 21395374/7747M out.
Disks: 17026780/555G read, 15757470/638G written.

PID    COMMAND      %CPU  TIME     #TH    #WQ  #PORTS MEM    PURG   CMPRS  PGRP  PPID  STATE
90498  bash         0.0   00:00.41 1      0    21     1080K  0B     564K   90498 90497 sleeping
90497  login        0.0   00:00.10 2      1    31     1236K  0B     1220K  90497 90496 sleeping
90496  Terminal     0.5   01:43.28 6      1    378-   103M-  16M    13M    90496 1     sleeping
89197  siriknowledg 0.0   00:00.83 2      2    45     2664K  0B     1528K  89197 1     sleeping
89193  com.apple.DF 1.0   00:17.34 2      1    68     2688K  0B     1700K  89193 1     sleeping
82655  LookupViewSe 0.0   00:10.75 3      1    169    13M    0B     8064K  82655 1     sleeping
82453  PAH_Extensio 0.1   00:25.89 3      1    235    15M    0B     7996K  82453 1     sleeping
75819  tzlinkd      0.0   00:00.01 2      2    13     452K   0B     444K   75819 1     sleeping
75787  MTLCompilerS 0.0   00:00.10 2      2    74     9032K  0B     9020K  75787 1     sleeping
75776  secd         0.0   00:00.78 2      2    36     3208K  0B     2328K  75776 1     sleeping
75098  DiskUnmountW 0.0   00:00.48 2      2    34     1420K  0B     728K   75098 1     sleeping
75093  MTLCompilerS 0.0   00:00.06 2      2    21     5924K  0B     5912K  75093 1     sleeping
74938  ssh-agent    0.0   00:00.00 1      0    21     908K   0B     892K   74938 1     sleeping
74063  Google Chrom 0.0   10:48.49 15     1    678    192M   0B     51M    54320 54320 sleeping
```

- Memory stays about 80% used
- A lot of it is shared 1.9 GB

Many Uses of Virtual Memory and “Demand Paging” …

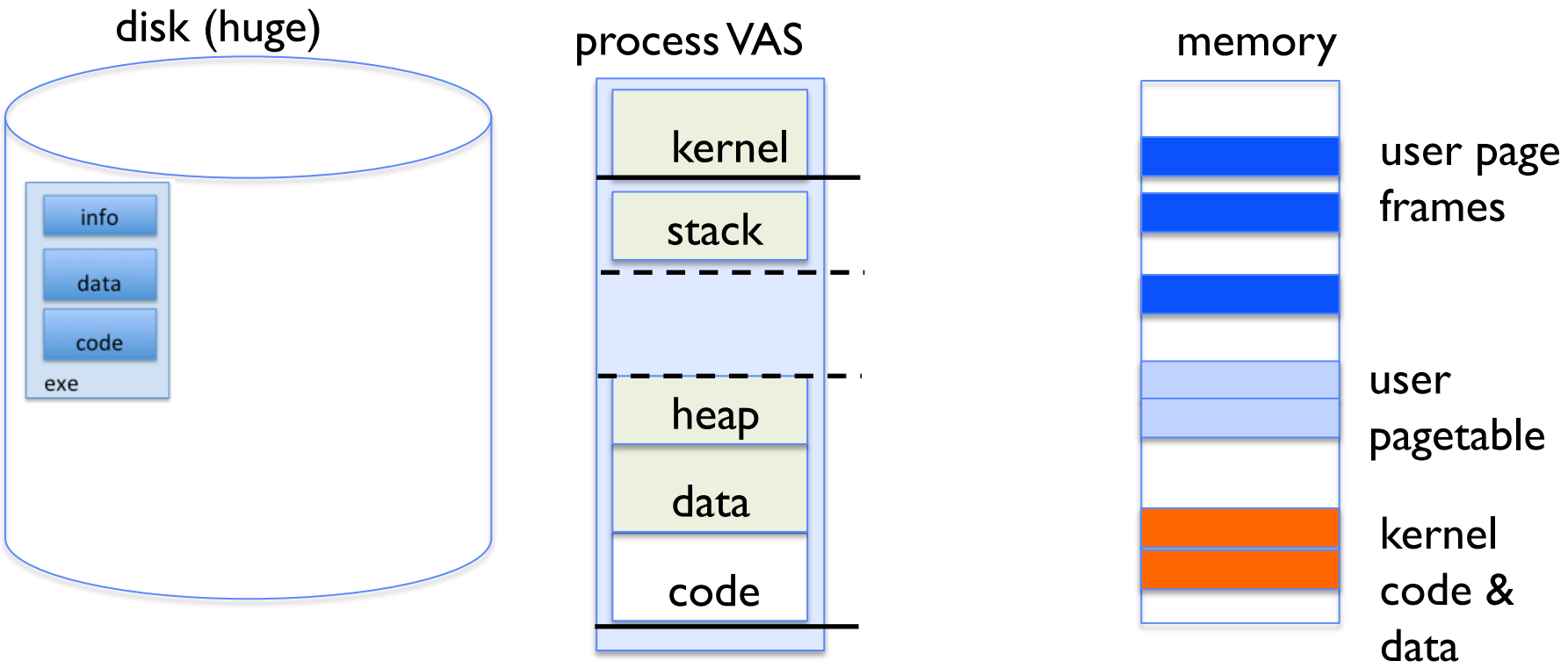
- Extend the stack
  - Allocate a page and zero it
- Extend the heap
- Process Fork
  - Create a copy of the page table
  - Entries refer to parent pages – NO-WRITE
  - Shared read-only pages remain shared
  - Copy page on write
- Exec
  - Only bring in parts of the binary in active use
  - Do this on demand
- MMAP to explicitly share region (or to access a file as RAM)

Classic: Loading an Executable into Memory



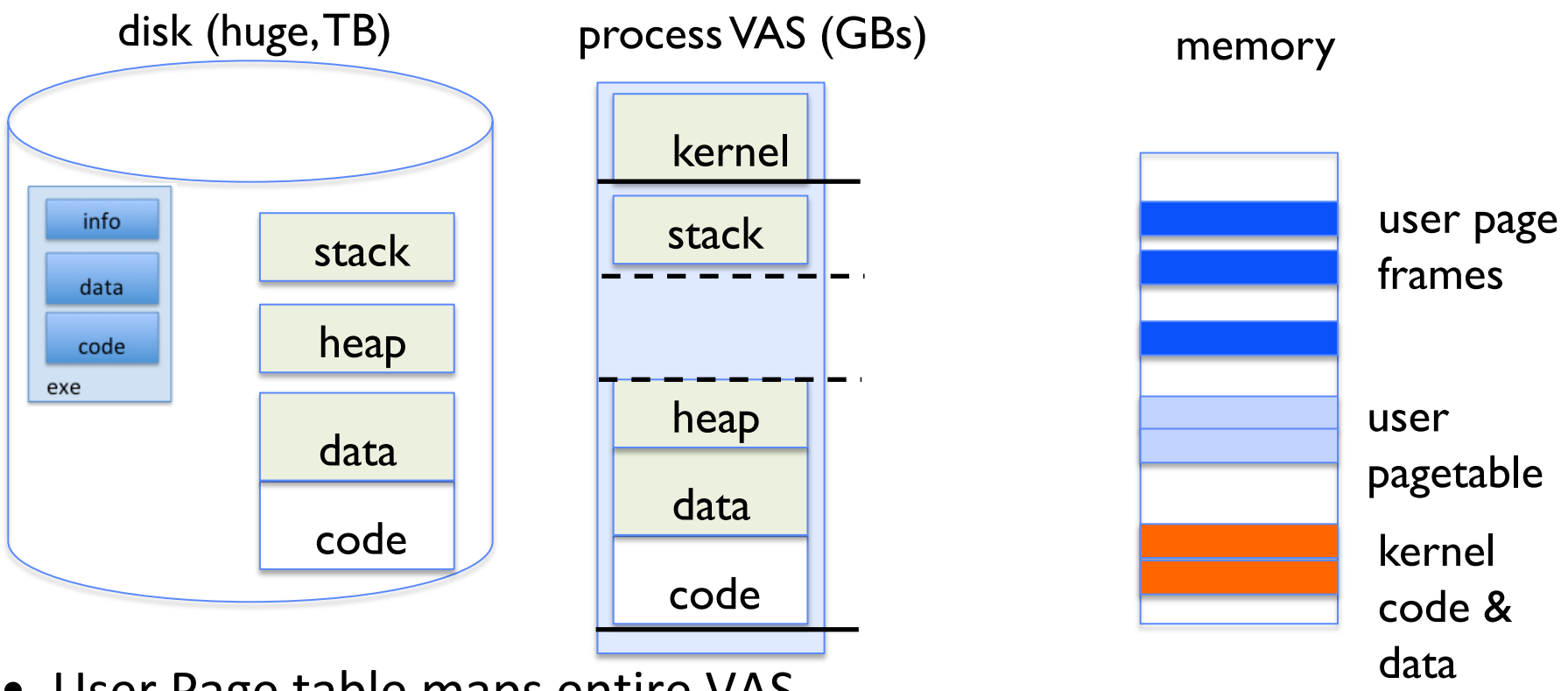
- `.exe`
  - lives on disk in the file system
  - contains contents of code & data segments, relocation entries and symbols
  - OS loads it into memory, initializes registers (and initial stack pointer)
  - program sets up stack and heap upon initialization

Create Virtual Address Space of the Process



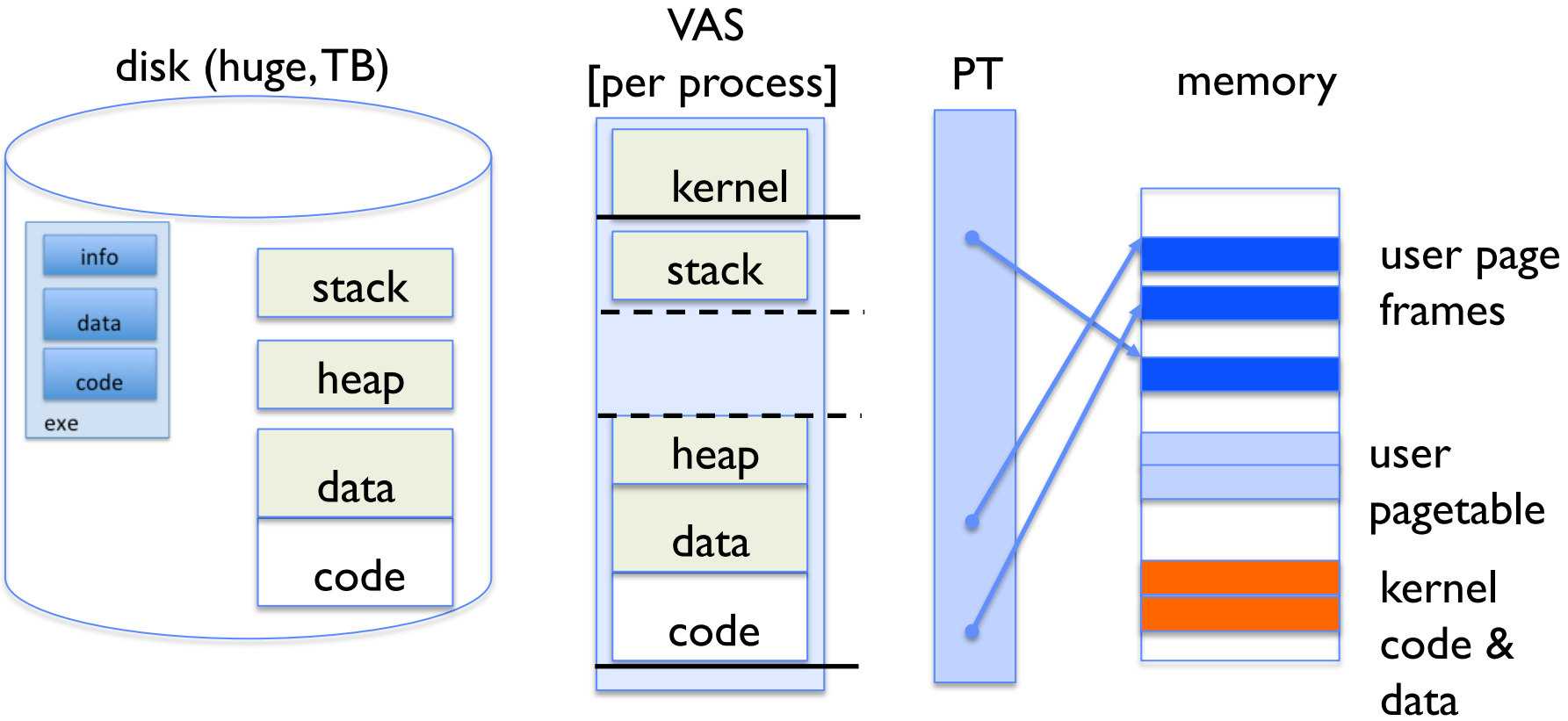
- Utilized pages in the VAS are backed by a page block on disk
  - Called the backing store or swap file
  - Typically, in an optimized block store, but can think of it like a file

Create Virtual Address Space of the Process



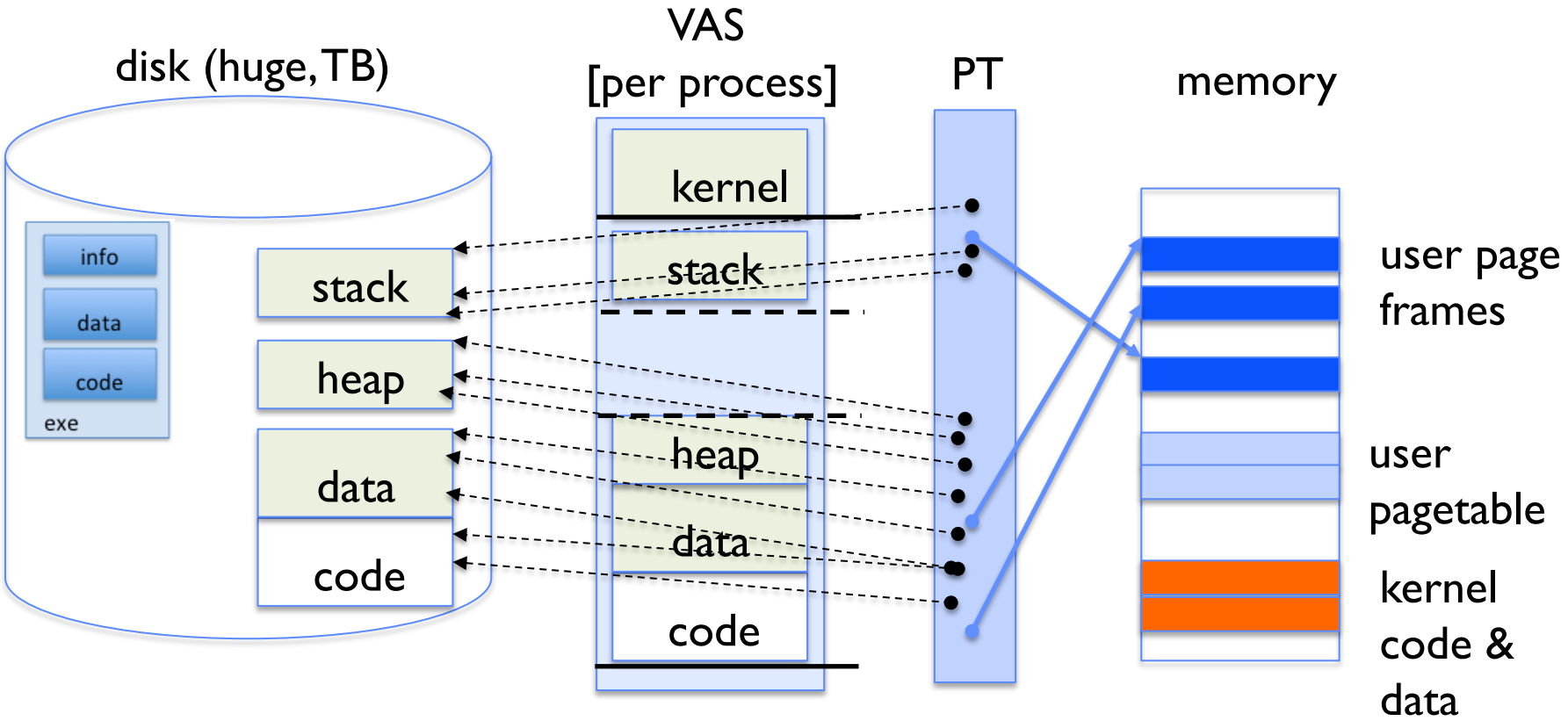
- User Page table maps entire VAS
- All the utilized regions are backed on disk
  - swapped into and out of memory as needed
- For *every* process

Create Virtual Address Space of the Process



- User Page table maps entire VAS
  - Resident pages mapped to the frame in memory they occupy
  - The portion of page table that the HW needs to access must be resident in memory

Provide Backing Store for VAS

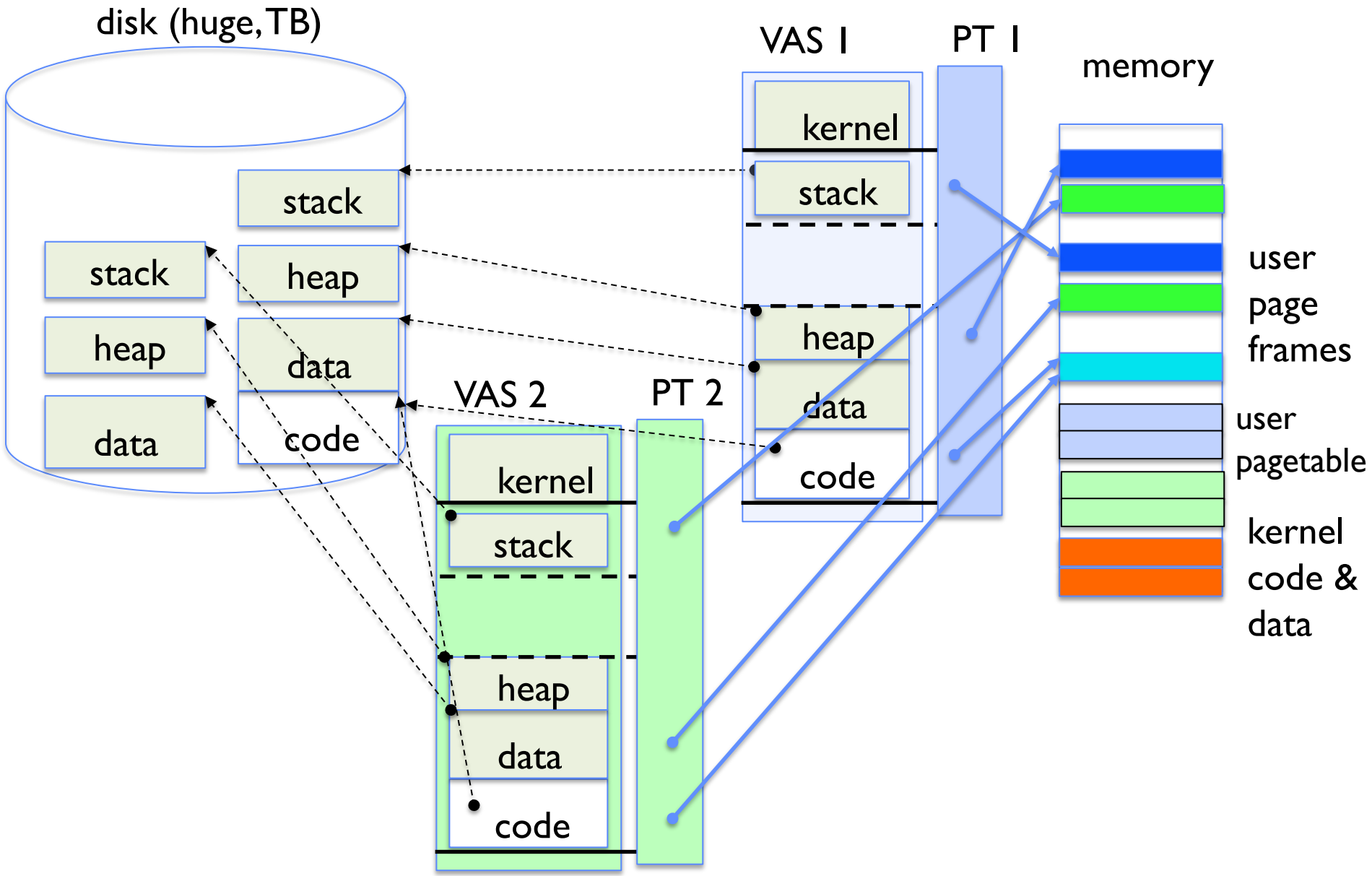


- User Page table maps entire VAS
- Resident pages mapped to the frame in memory they occupy
- For all other pages, OS must record where to find them on disk

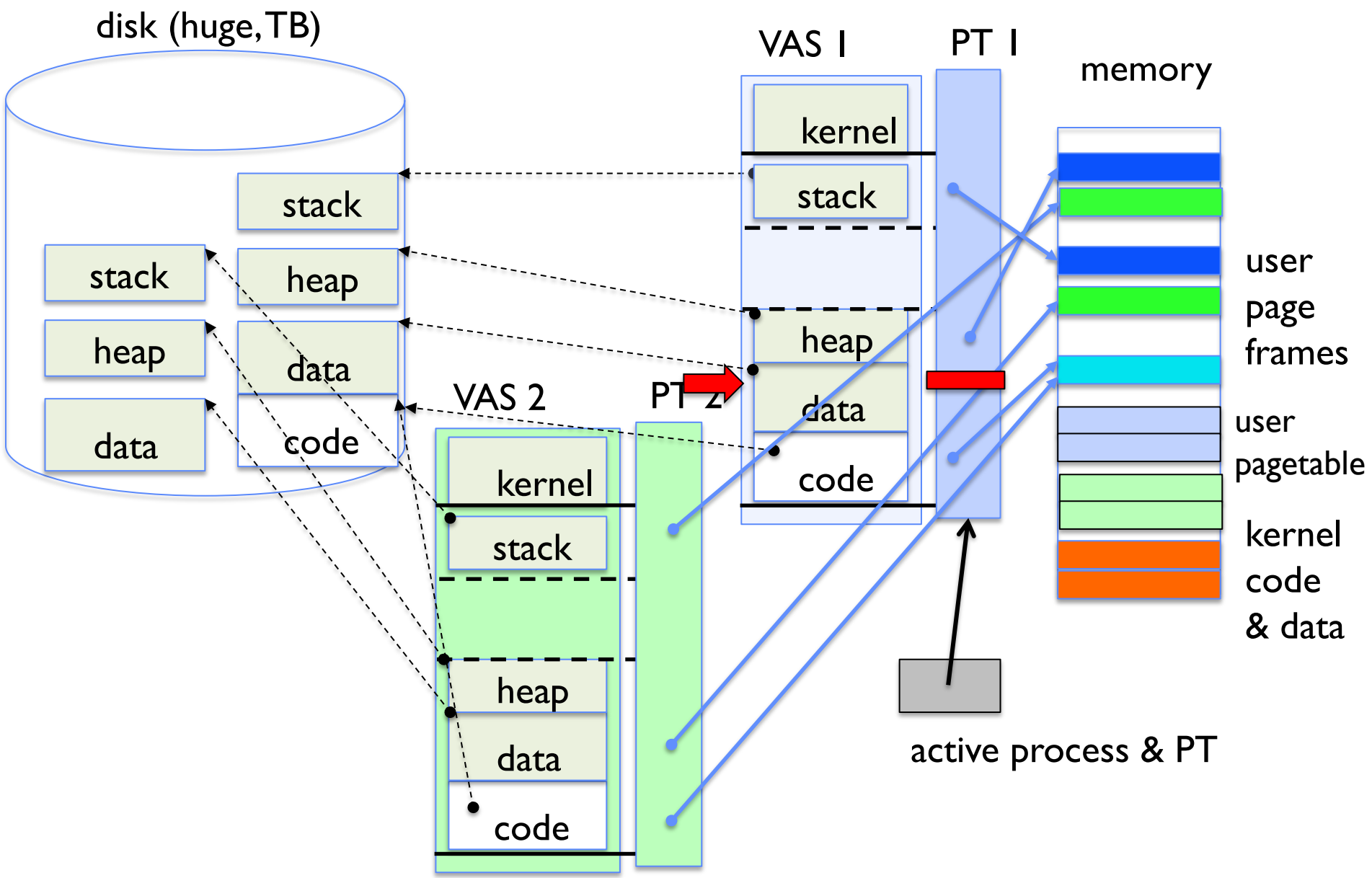
What Data Structure Maps Non-Resident Pages to Disk?

- `FindBlock(PID, page#)` → `disk_block`
  - Some OSs utilize spare space in PTE for paged blocks
  - Like the PT, but purely software
- Where to store it?
  - In memory – can be compact representation if swap storage is contiguous on disk
  - Could use hash table (like Inverted PT)
- Usually want backing store for resident pages too
- May map code segment directly to on-disk image
  - Saves a copy of code to swap file
- May share code segment with multiple instances of the program

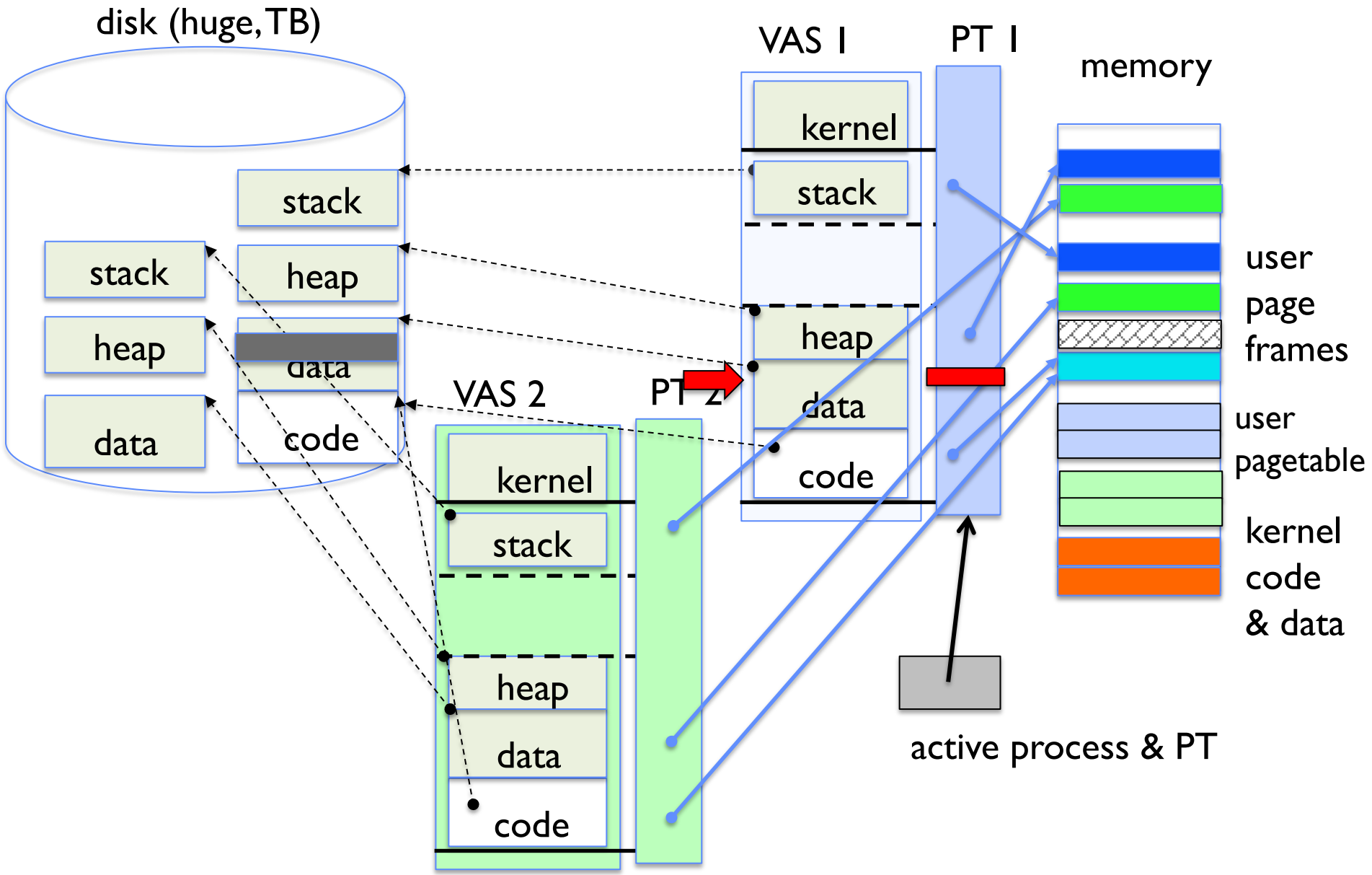
Provide Backing Store for VAS



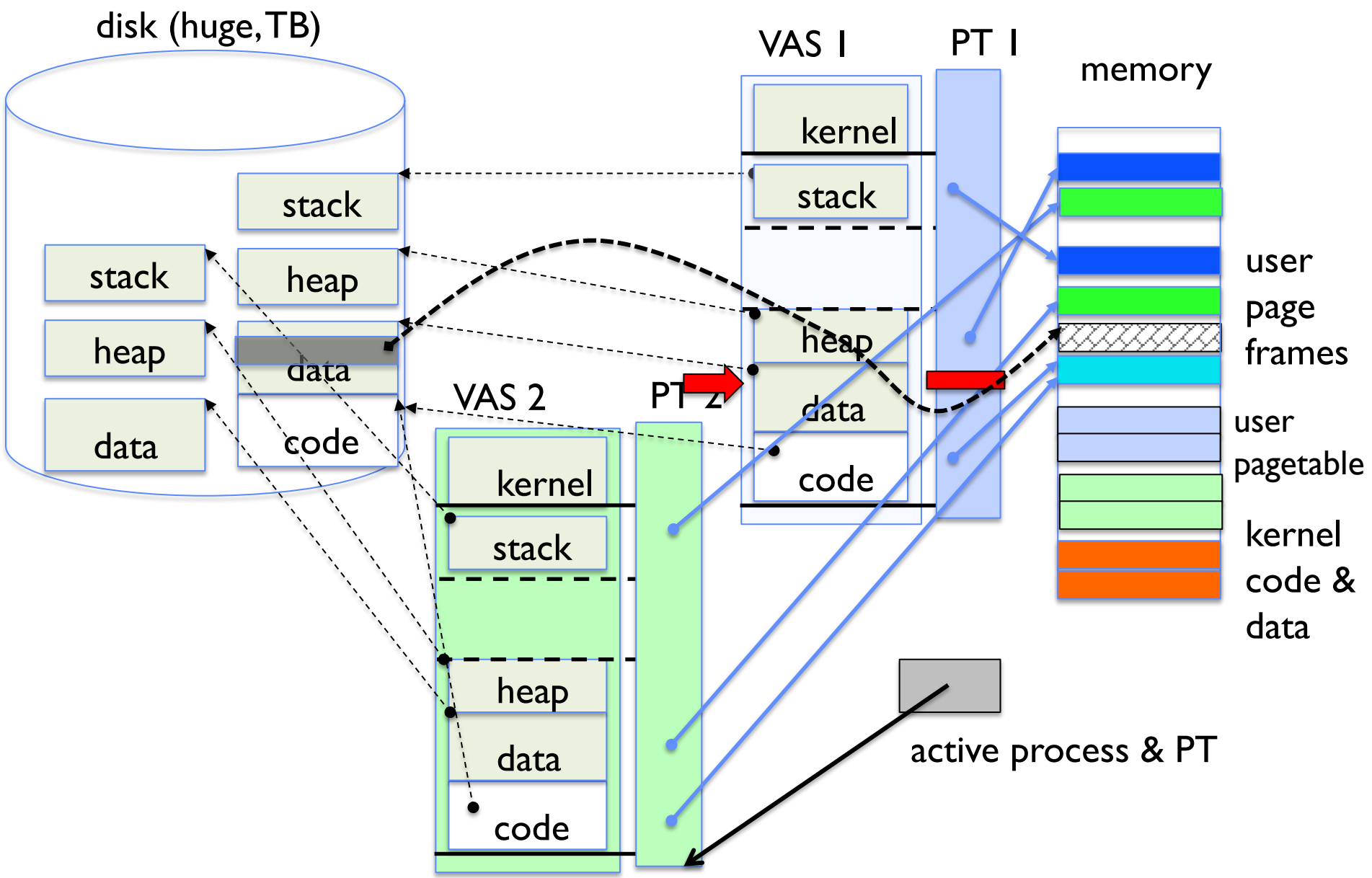
On Page Fault …



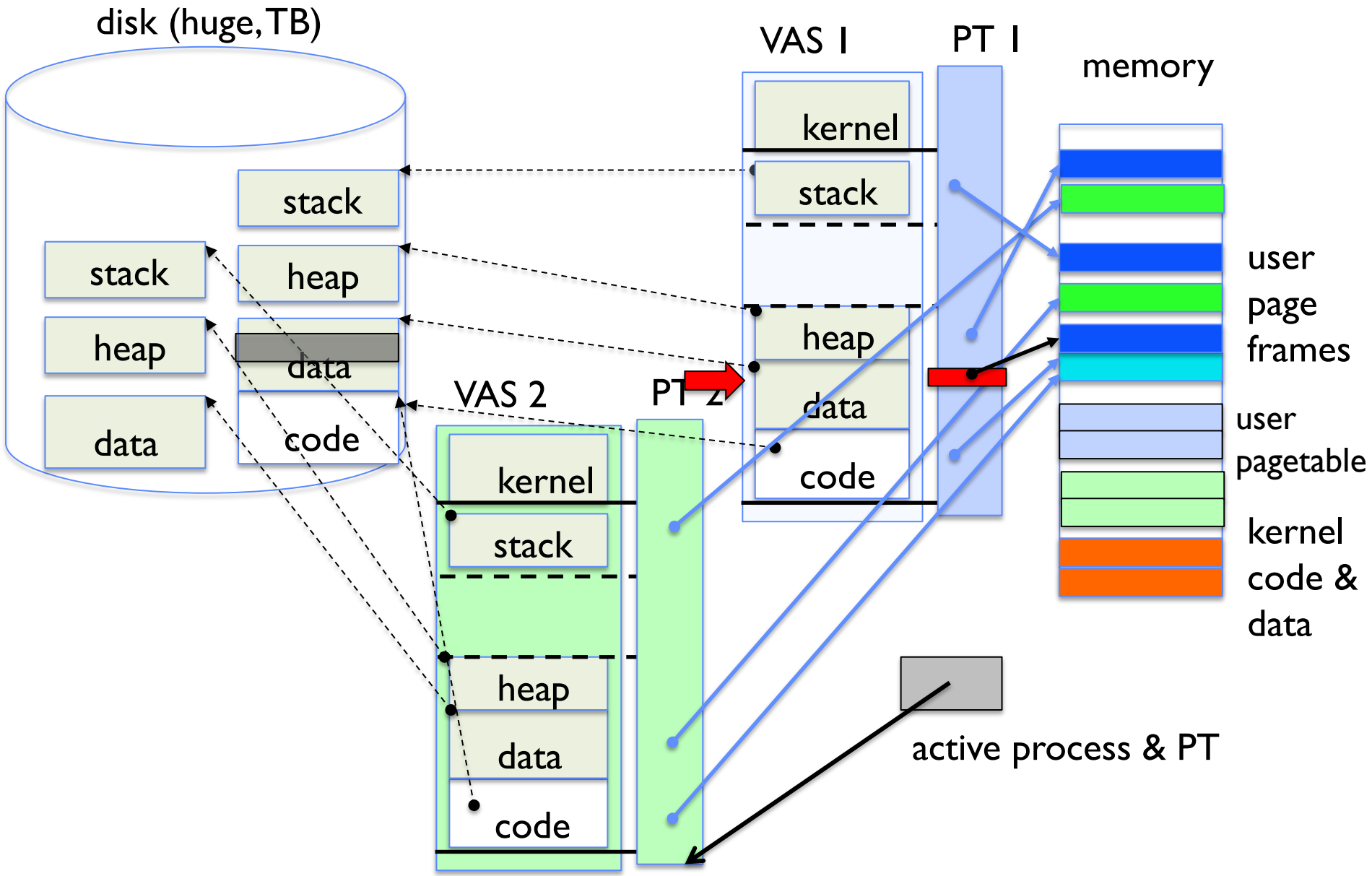
On Page Fault … Find & Start Load



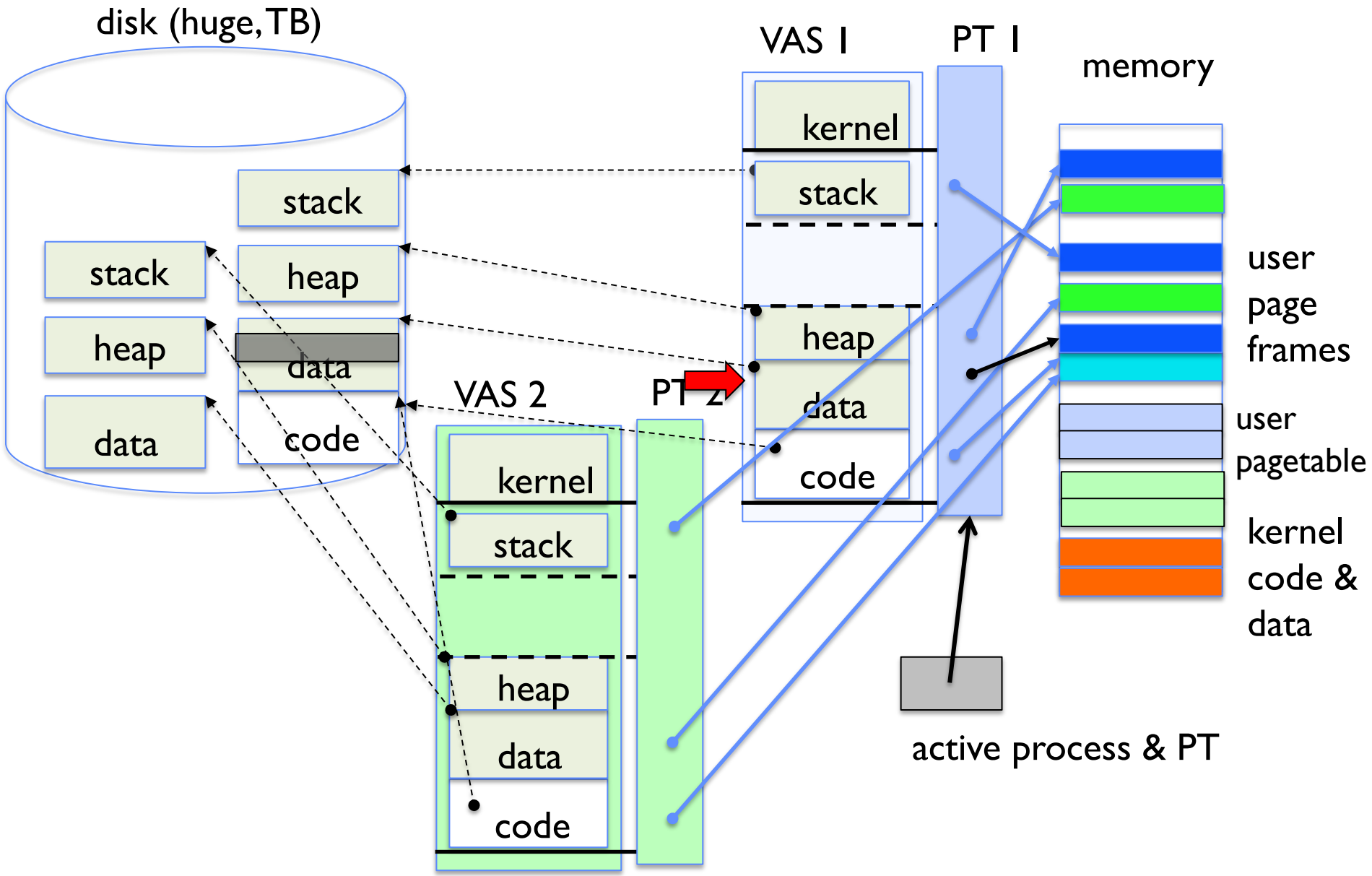
On page Fault … schedule other Process or Thread



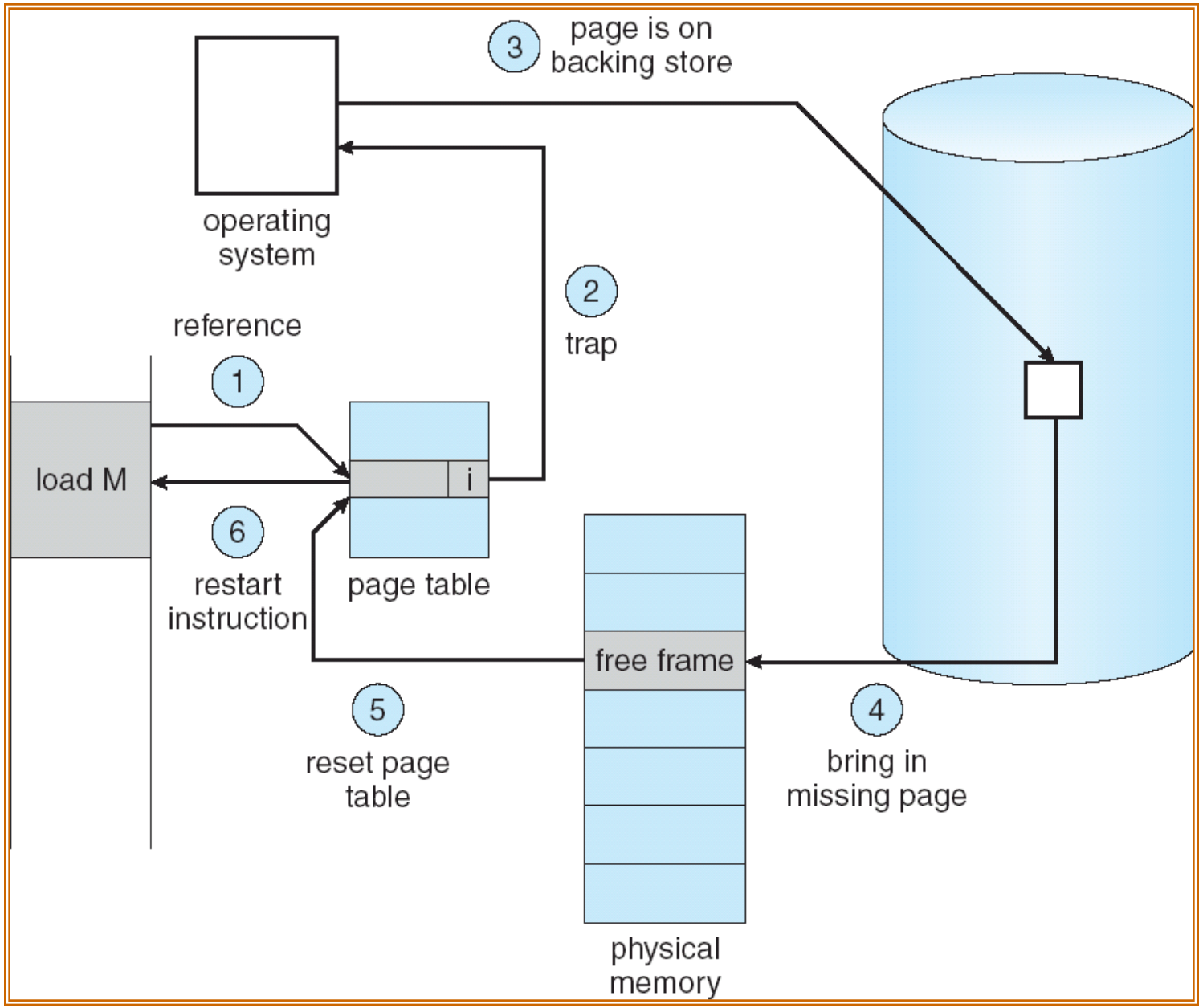
On Page Fault … Update PTE



Eventually reschedule faulting thread



Summary: Steps in Handling a Page Fault



Some questions we need to answer!

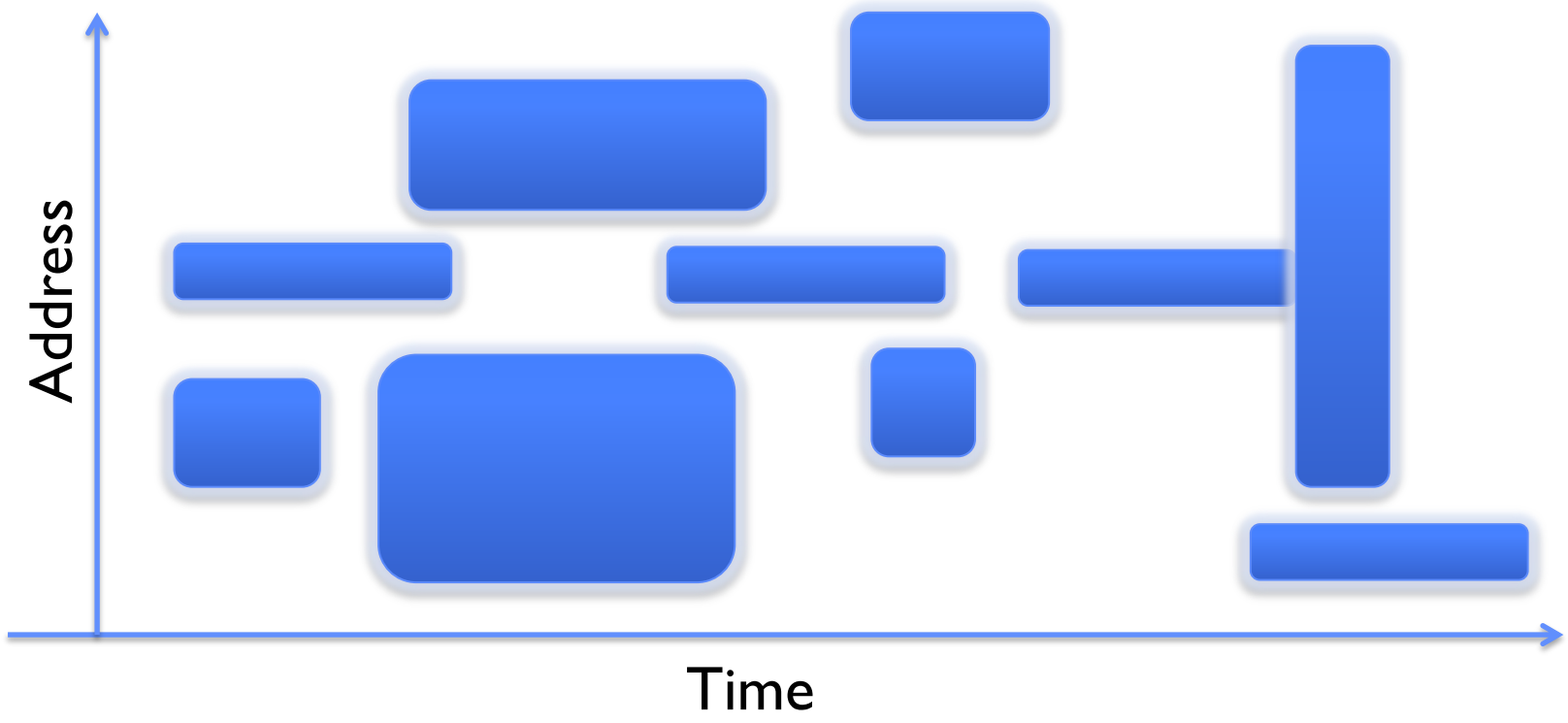
- During a page fault, where does the OS get a free frame?
  - Keeps a free list
  - Unix runs a “reaper” if memory gets too full
    - » Schedule dirty pages to be written back on disk
    - » Zero (clean) pages which haven’t been accessed in a while
  - As a last resort, evict a page first

- How can we organize these mechanisms?
  - Work on the replacement policy

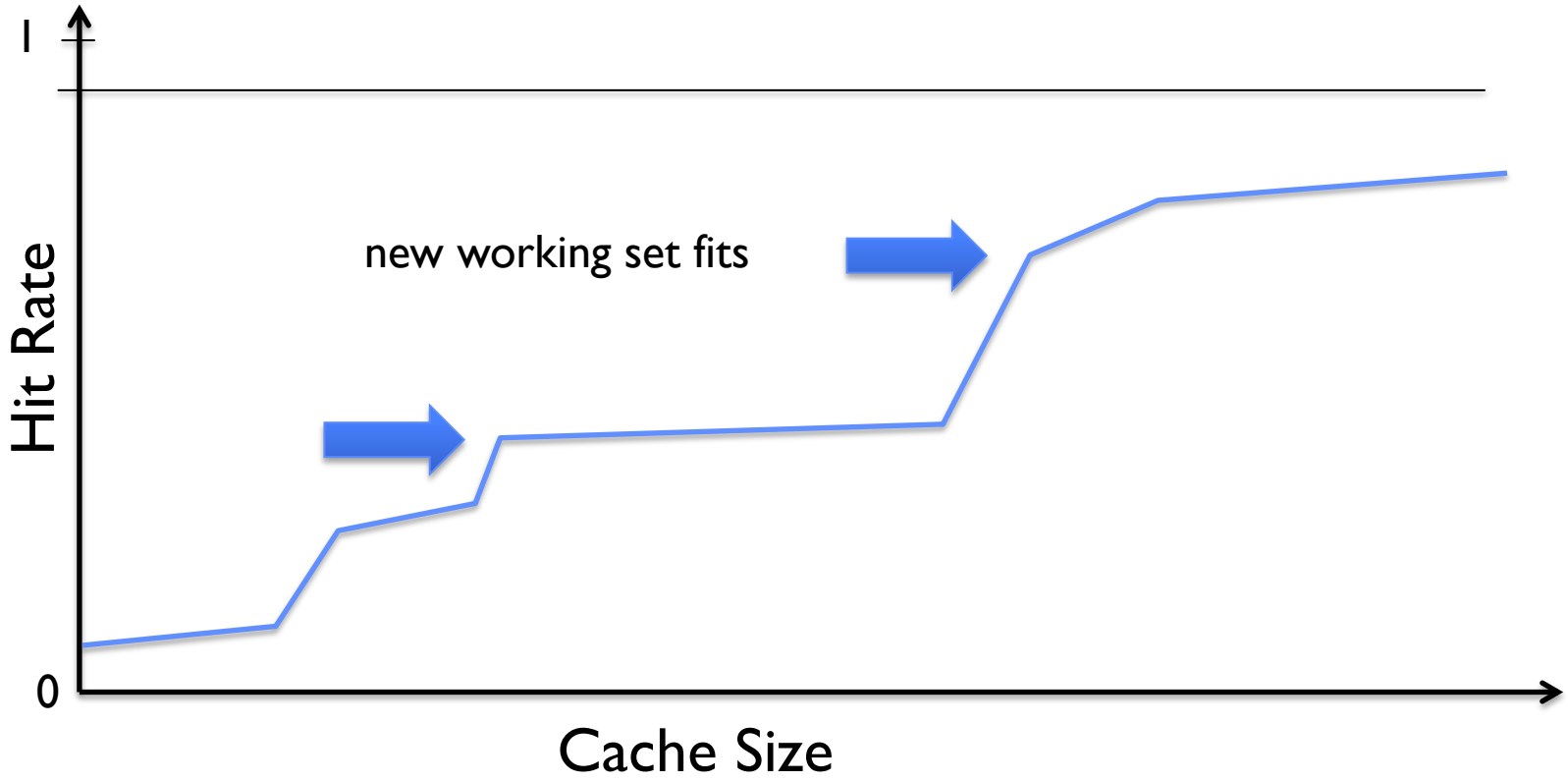
- How many page frames/process?
  - Like thread scheduling, need to “schedule” memory resources:
    - » Utilization? fairness? priority?
  - Allocation of disk paging bandwidth

Working Set Model

- As a program executes it transitions through a sequence of "working sets" consisting of varying sized subsets of the address space



Cache Behavior under WS model



- Amortized by fraction of time the Working Set is active
- Transitions from one WS to the next
- Capacity, Conflict, Compulsory misses
- Applicable to memory caches and pages

Demand Paging Cost Model

- Since Demand Paging like caching, can compute average access time! ("Effective Access Time")
  - EAT = Hit Rate x Hit Time + Miss Rate x Miss Time (Hit Rate + Miss Rate = 1)
  - EAT = Hit Time + Miss Rate x Miss Penalty (Miss Penalty = Miss Time – Hit Time)
- Example:
  - Memory access time = 200 nanoseconds
  - Average page-fault service time (Miss Penalty) = 8 milliseconds
  - Suppose p = Probability of miss, 1-p = Probably of hit
  - Then, we can compute EAT as follows:

    EAT = 200ns + p x 8 ms

    = 200ns + p x 8,000,000ns
- If one access out of 1,000 causes a page fault, then EAT = 8.2 µs:
  - This is a slowdown by a factor of 40x !
- What if want slowdown by less than 10%?
  - EAT < 200ns x 1.1 $\Rightarrow$ p < 2.5 x 10^{-6}
  - This is about 1 page fault in 400,000!

Group Discussion

- Compulsory Misses:
  - What are they in the context of demand paging?
  - How might we remove these misses?

- Capacity Misses:
  - What are they in the context of demand paging?
  - How might we remove these misses?

- Conflict Misses:
  - What are they in the context of demand paging?
  - How might we remove these misses?

- Policy Misses:
  - What are they in the context of demand paging?
  - How might we remove these misses?

What Factors Lead to Misses in Page Cache?

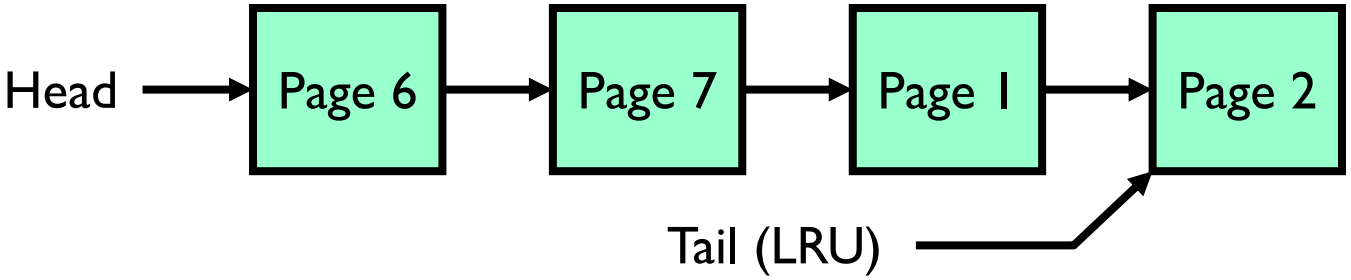
- Compulsory Misses:
  - Pages that have never been paged into memory before
  - How might we remove these misses?
    - » Prefetching: loading them into memory before needed
    - » Need to predict future somehow!
- Capacity Misses:
  - Not enough memory. Must somehow increase available memory size.
  - Can we do this?
    - » One option: Increase amount of DRAM (not quick fix!)
    - » Another option: If multiple processes in memory: adjust percentage of memory allocated to each one!
- Conflict Misses:
  - Technically, conflict misses don't exist in virtual memory, since it is a "fully-associative" cache
- Policy Misses:
  - Caused when pages were in memory, but kicked out prematurely because of the replacement policy
  - How to fix? Better replacement policy

Page Replacement Policies

- Why do we care about Replacement Policy?
  - Replacement is an issue with any cache
  - Particularly important with pages
    - » The cost of being wrong is high: must go to disk
    - » Must keep important pages in memory, not toss them out
- FIFO (First In, First Out)
  - Throw out oldest page. Be fair – let every page live in memory for same amount of time.
  - Bad – throws out heavily used pages instead of infrequently used
- RANDOM:
  - Pick random page for every replacement
  - Typical solution for TLB's. Simple hardware
  - Pretty unpredictable – makes it hard to make real-time guarantees
- MIN (Minimum):
  - Replace page that won't be used for the longest time
  - Great (provably optimal), but can't really know future…
  - But past is a good predictor of the future …

Replacement Policies (Con't)

- LRU (Least Recently Used):
  - Replace page that hasn't been used for the longest time
  - Programs have locality, so if something not used for a while, unlikely to be used in the near future.
  - Seems like LRU should be a good approximation to MIN.
- How to implement LRU? Use a list:



  - On each use, remove page from list and place at head
  - LRU page is at tail
- Problems with this scheme for paging?
  - Need to know immediately when page used so that can change position in list…
  - Many instructions for each hardware access
- In practice, people approximate LRU (more later)

Summary

- Demand Paging: Treating the DRAM as a cache on disk
  - Page table tracks which pages are in memory
  - Any attempt to access a page that is not in memory generates a page fault, which causes OS to bring missing page into memory
- Replacement policies
  - FIFO: Place pages on queue, replace page at end
  - RANDOM: Pick random page for every replacement
  - MIN: Replace page that will be used farthest in future
  - LRU: Replace page used farthest in past